

# Afillar Insurance APIs

## Developer Guide

*Architecture, code, configuration, workflow, portal and operations: the hand-over reference  
for maintainers*

Version 1.2

05-Oct-2026

Prepared for the Afillar Channel Integration Program

# Table of Contents

|                                                    |    |
|----------------------------------------------------|----|
| Table of Contents .....                            | 2  |
| 1. Introduction .....                              | 4  |
| 1.1 Purpose of this Guide .....                    | 4  |
| 1.2 What the Solution Is .....                     | 4  |
| 1.3 Glossary .....                                 | 4  |
| 1.4 Read This First: What Will Surprise You .....  | 5  |
| 1.5 Release History .....                          | 6  |
| 2. Architecture .....                              | 7  |
| 2.1 Layers .....                                   | 7  |
| 2.2 Design Rules .....                             | 7  |
| 2.3 Request Lifecycle .....                        | 8  |
| 3. Source Layout and Tooling .....                 | 9  |
| 3.1 Workspace Layout .....                         | 9  |
| 3.2 The imo Tool .....                             | 9  |
| 3.3 Groovy Rules of This Runtime .....             | 9  |
| 3.4 Deployment Order of the Code .....             | 10 |
| 4. Code Reference .....                            | 11 |
| 4.1 Asset Inventory .....                          | 11 |
| 4.2 Common Functions .....                         | 12 |
| 4.3 APIs .....                                     | 20 |
| 5. API Walkthroughs .....                          | 24 |
| 6. Data and Configuration Reference .....          | 26 |
| 6.1 Tables .....                                   | 26 |
| 6.2 Configuration Keys (UP_ApiConfig) .....        | 27 |
| 6.3 Field Rules (UP_FieldRule) .....               | 30 |
| 6.4 Security Baselines (UPSecurity) .....          | 31 |
| 7. Platform Configuration (Products) .....         | 32 |
| 7.1 What Exists on the Platform .....              | 32 |
| 7.2 Data Dictionary Bindings .....                 | 32 |
| 7.3 How the Products Were Built .....              | 32 |
| 8. Business Algorithms .....                       | 33 |
| 8.1 Premium Mode, Term and Age .....               | 33 |
| 8.2 Cancellation (Checker / Approver) .....        | 33 |
| 8.3 Platform Quirks That Shaped the Code .....     | 34 |
| 9. Errors, Logging and Tracing .....               | 35 |
| 10. How-To Recipes .....                           | 36 |
| 11. Test Tooling .....                             | 38 |
| 12. Test Procedure for a Change .....              | 41 |
| 13. Environments and Deployment .....              | 42 |
| 14. Known Limitations and Backlog .....            | 43 |
| 14.1 Troubleshooting .....                         | 43 |
| 14.2 First-Day Checklist for a New Developer ..... | 43 |
| 15. Workflow, Roles and Dashboard .....            | 44 |
| 15.1 Overview .....                                | 44 |
| 15.2 Roles and Identity .....                      | 44 |
| 15.3 The Maker Stamp (AgentCode) .....             | 45 |
| 15.4 Queues .....                                  | 45 |
| 15.5 Dashboard .....                               | 46 |
| 15.6 Maker-Checker Rule and the Test Switch .....  | 46 |

|                                                    |    |
|----------------------------------------------------|----|
| 15.7 New API Contracts (examples)                  | 46 |
| 16. Notifications (Email and SMS)                  | 48 |
| 16.1 Design                                        | 48 |
| 16.2 Events and Templates                          | 48 |
| 16.3 Modes and Result Status                       | 48 |
| 16.4 SNS Integration Details                       | 48 |
| 16.5 Switching to Live Delivery                    | 49 |
| 17. The Portal (UIC Page)                          | 50 |
| 17.1 What and Where                                | 50 |
| 17.2 Structure of the File                         | 50 |
| 17.3 Sign-In and Session                           | 52 |
| 17.4 API Access                                    | 52 |
| 17.5 Configuration and Branding                    | 53 |
| 17.6 Deploy and Test the Portal                    | 53 |
| 17.7 Portal Limitations                            | 53 |
| 18. PA001, iHub and the Other Assets in the Tenant | 54 |
| 18.1 What Exists                                   | 54 |
| 18.2 How the PA001 APIs Work                       | 54 |
| 18.3 Comparison                                    | 55 |
| 18.4 What Is Not Known                             | 55 |
| 18.5 If iHub Is Wanted Later                       | 55 |
| 19. Operations Runbook and Handover Checklist      | 56 |
| 19.1 Switches and How to Change Them               | 56 |
| 19.2 Users and Roles                               | 56 |
| 19.3 Troubleshooting the v1.2 Features             | 56 |
| 19.4 Go-Live Checklist for the v1.2 Additions      | 57 |
| 19.5 What a New Developer Needs                    | 57 |
| 19.6 Backlog (Consolidated)                        | 57 |
| Appendix A. File Inventory                         | 59 |
| Appendix B. Table Definitions                      | 60 |

# 1. Introduction

## 1.1 Purpose of this Guide

This guide is for the developers who will maintain and extend the Afillar Insurance Company insurance APIs after hand-over. It explains how the solution is built, where every piece of logic and data lives, how a request travels through the code, which platform services are called and why, which platform quirks shaped the design, how to change or add things safely, how to test, and how to deploy. The reference sections (4, 6, 11, 12 and the appendices) are generated from the real source files, tables and scripts, so they match what is deployed in the MC tenant at the time of writing. Version 1.2 adds the maker-checker workflow, the dashboard, notifications, the staff portal and an analysis of PA001 and iHub (sections 15 to 19). **Section 1.4 lists what is most likely to surprise a new developer: read it first.**

## 1.2 What the Solution Is

A set of 18 active REST APIs (and 2 inactive ones: a retired stub and a test probe) with 13 shared functions, written as iComposer Groovy scripts on the InsureMO platform, plus a staff portal (a UIC page). They orchestrate the platform's proposal, policy, endorsement, search and print services for seven insurance products (UPPA001 Personal Accident, TL001 Term Life, FP001 Female Protection, HC001 Hospital Cash, CI001 Critical Illness, LP001 Life Protect, DH001 Domestic Helper; 11 variants, 57 plans). Partners such as Sohar Bank call the APIs; the APIs validate the request, call the platform and return a stable contract. Bank staff work in three roles (Maker, Proposal Checker, Cancellation Checker) through the portal. Everything that a business user may want to change (validation rules, product data, refund rules, texts, formats, channels, insurers, users and roles, message texts) is data in iTables, not code. The solution is developed in iComposer only: iHub is not used (section 18).

## 1.3 Glossary

| Term                          | Meaning                                                                                                                                                                        |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MC                            | The current development tenant in which everything is built and tested (tenant code <code>uponly</code> , gateway portal-gw.insuremo.com). UAT is a separate tenant.           |
| iComposer                     | InsureMO's low-code Groovy runtime. An API script is exposed as a REST route; a function (common service) is a shared Groovy class called with <code>getCommonService</code> . |
| iTables / data table          | Tenant configuration tables read by the code ( <code>UP_ApiConfig</code> , <code>UP_FieldRule</code> , ...).                                                                   |
| Tech product / market product | The platform keeps a technical product ( <code>&lt;CODE&gt;_TECH</code> , holds the data dictionary tree) and a market product ( <code>&lt;CODE&gt;</code> ) per product code. |
| DD / binding                  | Data dictionary: the fields of the policy tree. A field must be bound to an object before the platform accepts or calculates it.                                               |
| Maker                         | Bank employee who creates quotations and proposals and raises cancellation requests.                                                                                           |
| Proposal Checker              | Bank employee who approves (issues) or rejects a submitted proposal.                                                                                                           |
| Cancellation Checker          | Bank employee who approves or rejects a cancellation request (named Approver in v1.0 of the cancellation API).                                                                 |
| AgentCode                     | Platform proposal field in which the maker's platform user id is stored (section 15.3).                                                                                        |
| SNS                           | The platform's notification service used to send email and SMS (section 16).                                                                                                   |
| UIC                           | InsureMO UI Connector: hosts the portal as a single-page web app (section 17).                                                                                                 |
| Dry run                       | Notification mode in which messages are recorded in the response but not sent.                                                                                                 |
| ChannelCode                   | The partner that owns a record (SOHAR = Sohar Bank); sent on every request and saved on every proposal.                                                                        |
| OrgCode                       | Issuing branch code; the platform validates it against the code table PubBranch (value 1).                                                                                     |
| Carrier / insurer             | The insurer of a product: <code>CarrierCode</code> AFIC, name from table <code>UP_Carrier</code> .                                                                             |
| Trace id                      | Platform request id, returned as <code>trace_id</code> and used to find the log of a call.                                                                                     |

## 1.4 Read This First: What Will Surprise You

These are the facts most likely to cost a new developer a day. Each one is explained in the section named in the last column.

| #  | What                                                                                                                                                                                                                                                                                                                                                                        | Why it matters                                                                                                                                                                                                                     | Section |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1  | The MC tenant runs with <code>RoleEnforcement = Y</code> , <code>AllowSelfApproval = Y</code> and <code>NotifyMode = DRYRUN</code> . <code>UP_UserRole</code> holds three <b>test</b> rows for one account (ravi.teja@insuremo.com holds all three roles).                                                                                                                  | One account can run the whole flow, so maker and checker are the same person. Before UAT or go-live: <code>AllowSelfApproval = N</code> , replace the test rows with bank staff.                                                   | 15, 19  |
| 2  | The maker's platform user id is stored in the proposal field <b>AgentCode</b> . Queues, "My submissions" and the maker-checker rule all depend on it.                                                                                                                                                                                                                       | <code>AgentCode</code> is a platform field that normally means a sales agent. It is the only searchable field available. It is in the server-owned list, so callers cannot set it. Proposals created before v1.2 have no maker id. | 15.3    |
| 3  | The work queues and dashboard counts are <b>read from the platform search index</b> , not from our tables. The index limit is 1,000 rows per query, it has no channel, no maker (except <code>AgentCode</code> ), an empty insured name and a zero premium.                                                                                                                 | Every queue row is enriched by loading the proposal (one call per row). Page size x page number must not exceed 1,000. A date window is always applied.                                                                            | 15.4    |
| 4  | Dashboard premium and commission are calculated by <b>loading policies</b> , capped at <code>DashboardMaxPolicies</code> (60) newest per product and flagged <code>Truncated</code> .                                                                                                                                                                                       | Totals are exact only while a product has no more policies than the cap in the window. The dashboard takes 7 to 10 seconds. It is not a reporting database.                                                                        | 15.5    |
| 5  | The MC tenant holds a lot of <b>test data</b> : several hundred pending test proposals and over a hundred pending test cancellations created by the test suites.                                                                                                                                                                                                            | The queues and dashboard show them. There is no API to delete them.                                                                                                                                                                | 19      |
| 6  | <b>No email or SMS is delivered.</b> The MC tenant has zero SNS email and SMS accounts. The SNS SMS request carries no free text, only a template code configured in the SNS account.                                                                                                                                                                                       | <code>NotifyMode = LIVE</code> has never been exercised. The code path compiles but is unproven (field names, template codes).                                                                                                     | 16      |
| 7  | The portal's <b>sign-in screen</b> posts the user name and password to the platform's CAS ticket service and takes the token from the reply. Only the error path and the "use my InsureMO session" path were tested. A successful password sign-in was never run (no password available).                                                                                   | The reply field name is assumed ( <code>access_token</code> / <code>token</code> / <code>ticket</code> , as in the Postman collection). Verify with a real login first.                                                            | 17.3    |
| 8  | The portal builds API URLs with the standard UIC helper ( <code>/{{tenant_code}}/api-orchestration-test/v1/flow</code> ). All automated tests ran with an override ( <code>sohar_api_base</code> in <code>sessionStorage</code> ) that points the page at the gateway through a stub. <b>The URL construction inside the real portal frame was not exercised by a test.</b> | If the first call in the real portal fails with 404, check the route prefix. The override hook should be removed before go-live.                                                                                                   | 17.4    |
| 9  | The <code>imo login</code> ( <code>imo auth login</code> ) lasts 24 hours and cannot be refreshed from a script. Every tool in the build folder needs a valid profile token.                                                                                                                                                                                                | When a push or script says "token is invalid or expired", log in again; nothing is wrong with the code.                                                                                                                            | 19      |
| 10 | PA001 and its APIs are a <b>different design</b> : iComposer APIs in group <code>Policy_API</code> that call an adapter ( <code>IhubCallerIntegrationAdapter</code> ) driven by the table <code>AdapterServiceCfg</code> . The new <code>/up</code> APIs do not use that adapter or iHub.                                                                                   | Do not reuse PA001 helper functions or tables for the new products. PA001 must stay untouched ( <code>snapshot.py</code> ).                                                                                                        | 18      |

| #  | What                                                                                                                                                                                                                                                                                                                                                                   | Why it matters                                                                                                                                                         | Section  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11 | Cancellation requests that are rejected are indistinguishable from requests replaced when an approval runs: both end as endorsement status 600 (withdrawn).                                                                                                                                                                                                            | The cancellation queue can only list PENDING (120) and ISSUED (300).                                                                                                   | 15.4     |
| 12 | The specification (v1.1), UAT plan, policy model, OpenAPI file and Postman collection were <b>not updated</b> for v1.2. This guide is the only document that describes the new APIs.                                                                                                                                                                                   | Section 15.7 holds request and response examples. Update the other documents before they are given to partners.                                                        | 15.7, 19 |
| 13 | GET /up/me omits the Roles list when the caller has no roles (the platform drops empty lists).                                                                                                                                                                                                                                                                         | Treat a missing Roles as an empty list (the portal does).                                                                                                              | 15.2     |
| 14 | An inactive probe API UPspikeWho (path /up/spike-who) remains in MC from the identity spike. The imo tool cannot delete an API.                                                                                                                                                                                                                                        | Delete it in the iComposer console before UAT. It must not be deployed.                                                                                                | 19       |
| 16 | <b>Get Plans runs the platform's quotation validation first</b> (as the PA001 Get Plans does), switched by GetPlansPlatformValidate (MC: Y). It rejects code values outside their code table (gender, residency), an unknown risk element and non-numeric salary, but accepts a wrong ID type and marital status code (neither the platform nor our rules check them). | A quote that was accepted before can now be refused with 400 UP-PLATFORM-VALIDATION or the platform's own 400. Set the key to N to return to the table-only behaviour. | 5, 9     |
| 15 | Static type checking on push: UPShaper.shapeResponse returns Object (cast it), SMS and email builders take requestBody(...), an anonymous Comparator may not use method variables.                                                                                                                                                                                     | Compile errors on push usually mean one of these; the fix is a cast or a different method name.                                                                        | 3.3      |

## 1.5 Release History

| Version | What it added                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 13 APIs for 7 products (get plans, proposal, update, issue, load, reject, documents, cancel check and approve, master data, commission, feed file), table-driven validation, security controls, full test suite, collections.                                                                                                                                                                                                                                                                                   |
| 1.1     | Insurer (CarrierCode / CarrierName, table UP_Carrier) in every response and an optional validated request field; developer guide and UAT plan.                                                                                                                                                                                                                                                                                                                                                                  |
| 1.2     | Roles (Maker, Proposal Checker, Cancellation Checker) with maker-checker rules; work queues, cancellation detail, dashboard, share by email / SMS and automatic alerts; the staff portal (UIC page) with sign-in; 5 new APIs, 3 new functions, 2 new tables, about 33 new configuration keys and 29 new field rules; analysis of PA001 and iHub. Update 5 Oct: Get Plans now runs the platform quotation validation like PA001 (key GetPlansPlatformValidate) and platform policy-validation errors return 400. |

## 2. Architecture

### 2.1 Layers

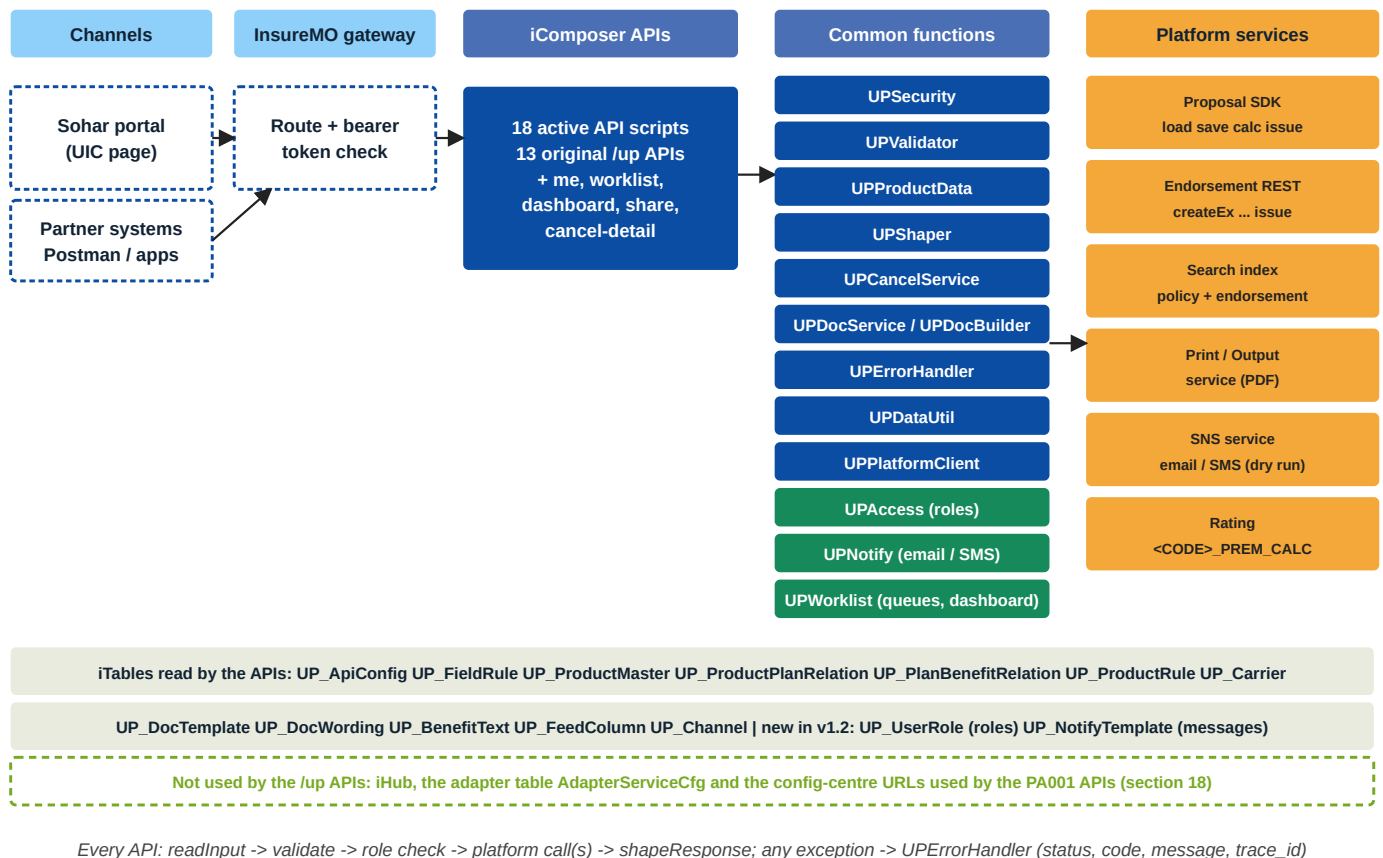


Figure 1: Architecture (v1.2) — channels, gateway, iComposer APIs, common functions, platform services and tables

Every API follows the same pipeline. The API script is thin: it reads the body through `UPSecurity.readInput`, validates it with `UPValidator.validate` (rules in table `UP_FieldRule` plus product and cross-field rules in code), calls the platform through the SDK clients (`ProposalSdkClient`) or the REST helper (`UPPlatformClient`), shapes the answer with `UPShaper.shapeResponse` and returns it. Any exception is turned into the standard error body by `UPErrorHandler.handle`, which also sets the HTTP status. The tables are read-only from the code (the `TablesSdkClient` is read-only); they are changed with the `imo` tool by an administrator. From v1.2 the API also checks the caller's role (`UPAccess`) and the platform search index is read for queues and the dashboard (`UPWorklist`).

### 2.2 Design Rules

- Everything configurable lives in tables: a missing configuration key stops the request with `UP-CONFIG` (HTTP 500, generic message, detail in the log); there is no silent default. The only exceptions are the security baselines in `UPSecurity`, which have an optional table override.
- PA001 (the original product, config and code) is never read or changed by this solution: product data lives in separate `UP_` tables and the new products are new platform products. A snapshot tool (`snapshot.py`) proves this before and after every change.
- The caller's own bearer token is used for every platform call; there is no service account. The token is only sent to hosts listed in `UP_ApiConfig.AllowedHosts` (https only).
- Responses are shaped for the channel: platform technical fields may be present but are not part of the contract.
- One error contract on every API: `status`, `code`, `message`, `errors[]`, `api`, `trace_id`. A missing role or a maker-checker breach is 403 `UP-403`.
- Roles come from the table `UP_UserRole` and are checked in the APIs, never in the portal alone. The portal menu is a convenience.

- No workflow state is kept in our own tables: proposals and cancellation requests are read from the platform (section 15).

## 2.3 Request Lifecycle

1. Gateway authenticates the bearer token (HTTP 401 on failure) and routes `/up/<path>` to the iComposer API.
2. API script: `UPSecurity.readInput(RequestBody())` — body must be a JSON object, then every string is scanned for blocked characters (400 `INVALID_BODY` / `INVALID_CHARACTER`).
3. `UPValidator.validate(api, input)` — rules from `UP_FieldRule` for this API (required, type, length, pattern, range, enum, code table, conditional `WhenPath`), then product rules (age, term, premium mode, plan availability), `CarrierCode` match, beneficiary total. All problems are returned together as 400 `UP-VALIDATION`.
4. Role check: `UPAccess.require(...)` (403 `UP-403`), then for approvals the maker-checker rule. State and ownership checks for existing records: record exists (404), belongs to the channel (`requireChannel`), product matches, proposal still open (`requireOpenProposal`, 409).
5. Platform call(s): `ProposalSdkClient` (application, load, save, `persistCalculate`, issue, `queryPolicy`) or `UPPlatformClient` (endorsement REST) or the print service.
6. `UPShaper.shapeResponse`: commission switch, premium fields, payment info restored, `CarrierCode` / `CarrierName` added. Then, where the action triggers one, `UPNotify.fire` adds `Notifications[]`.
7. Return the map (HTTP 200) or, on any exception, `UPErrorHandler.handle(apiName, ex)`.

## 3. Source Layout and Tooling

### 3.1 Workspace Layout

```
ic/
  src/dev/Tenant/UP_PRODUCTS_API/
    api/<Name>/<Name>.groovy      20 API scripts (18 active; UPCancelPolicy is a retired stub and UPSpikeWho
a probe, both inactive)
    function/<Name>/<Name>.groovy  13 common functions
    .metadata/icomposer/{api,function}/<Name>.metadata.json  per-profile server metadata (ids, version, md5,
path, method)
uponly-build/
  tables/                        build, test and report tooling (section 11)
                                the rows written to the UP_ tables (r_*.json), definitions (def_*.json),
local copies (rows_*.json)
  samples/                      recorded request / response of every flow step
  tests/ security/ carrier/ workflow/ test cases and results
uic_pages/uponly/portal/sohar-insurance-portal/  the portal (index.html, metadata.json, prompt.md) in the git
repository
deliverables/workflow_src/      v1.2 iComposer source, set-up and test scripts (git copy)
```

Module id 70, group id 1188491119; functions have scope `public` and status 1. APIs are POST except `UPLoadPolicy` and `UPMe` (GET). All paths are `/up/<name>`: `get-plans`, `proposal`, `update-proposal`, `issue`, `load-policy`, `proposal-reject`, `doc_generation`, `application_doc_generation`, `cancel-check`, `cancel-approve`, `list-master-table`, `commission-query`, `generate-feedfile`, and since v1.2 me, `worklist` (`UPWorklistApi`), `dashboard`, `share`, `cancel-detail` (the inactive ones are `cancel-policy` and `spike-who`).

### 3.2 The imo Tool

| Task                                         | Command (profile <code>portal:uponly</code> = MC)                                                                                            |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Push one file (compiles on the server first) | <code>imo icomposer push current &lt;file&gt; --profile portal:uponly</code>                                                                 |
| Preview a push                               | <code>... push current &lt;file&gt; --dry-run</code>                                                                                         |
| Create a function / API                      | <code>imo icomposer create function --name X --module-id 70 --group-id 1188491119 --status 1 --func-scope 1 --profile portal:uponly</code>   |
| Pull the server copy                         | <code>imo icomposer reload ...</code> ; on a conflict choose <code>--prefer-local</code> or <code>--prefer-server</code>                     |
| Read a table / rows                          | <code>imo config datatable by-name --name UP_ApiConfig --json, ... record-list --id &lt;tableId&gt; --json</code>                            |
| Add rows                                     | <code>imo config datatable record-batch-save --file rows.json --confirm</code> (insert only; the primary key <code>Id</code> must be unique) |
| Change one row                               | <code>imo config datatable record-save --file row.json --confirm</code> (include <code>RecordId</code> )                                     |
| Create a table                               | <code>imo config datatable create --file def.json --confirm</code>                                                                           |
| Find a call in the log                       | <code>imo log query</code> with the trace id from the response (section 9)                                                                   |

### 3.3 Groovy Rules of This Runtime

- The server type-checks Groovy statically on push. `@groovy.transform.Field` is not allowed and script-level variables are not visible inside methods: use a method that returns the constant (see `UPSecurity.blockedBaseline()`).
- `log` is not declared: use `org.slf4j.LoggerFactory.getLogger("Name")` (see `UPErrorHandler`).
- Shared per-request state goes into the request attributes (`RequestContextHolder`), as the table cache in `UPDataUtil.getRecords` does.
- A function that calls another one declares it with `(UPOther) getCommonService("UPOther")`; push the called function first when it is new.
- The platform REST client raises its own exception type; `UPPlatformClient` catches it and returns `{status, body}` so that callers decide.
- `UPShaper.shapeResponse` returns `Object`: cast it (`((Map<String, Object>) shaper.shapeResponse(r))` before adding fields. The SNS email and SMS builders take `requestBody(...)`; an anonymous `Comparator` must

- not use method variables; the accessor of a hyphenated SDK tag is camel case (`emailAccountApi()`).
- Empty lists are dropped from the JSON that the platform returns (for example `Roles` in `/up/me`): treat a missing list as empty.
- Keep each function focused; the largest file is `UPValidator`.

### 3.4 Deployment Order of the Code

Dependencies run from the bottom up. Push in this order (the same order is used for UAT): `UPDataUtil`, `UPSecurity`, `UPErrorHandler`, `UPPlatformClient`, `UPValidator`, `UPProductData`, `UPShaper`, `UPDocBuilder`, `UPDocService`, `UPAccess`, `UPNotify`, `UPWorklist`, `UPCancelService`, then the 18 active APIs. The tables `UP_UserRole` and `UP_NotifyTemplate` and the new configuration keys must exist before the new code runs. Pushing a caller before a new callee fails the compile check.

## 4. Code Reference

### 4.1 Asset Inventory

| Name                       | Kind | Route                               | Status   | Lines | Server version | Calls (common functions)                                                                           |
|----------------------------|------|-------------------------------------|----------|-------|----------------|----------------------------------------------------------------------------------------------------|
| UPApplicationDocGeneration | API  | POST /up/application_doc_generation | active   | 25    | 8              | UPAccess, UPDataUtil, UPDocService, UPErrorHandler, UPSecurity, UPValidator                        |
| UPCancelApprove            | API  | POST /up/cancel-approve             | active   | 25    | 6              | UPAccess, UPCancelService, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator                     |
| UPCancelCheck              | API  | POST /up/cancel-check               | active   | 26    | 6              | UPAccess, UPCancelService, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator                     |
| UPCancelDetail             | API  | POST /up/cancel-detail              | active   | 28    | 2              | UPAccess, UPCancelService, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator                     |
| UPCancelPolicy             | API  | POST /up/cancel-policy              | inactive | 5     | 6              |                                                                                                    |
| UPCommissionQuery          | API  | POST /up/commission-query           | active   | 137   | 11             | UPAccess, UPDataUtil, UPDocBuilder, UPErrorHandler, UPProductData, UPSecurity, UPValidator         |
| UPDashboard                | API  | POST /up/dashboard                  | active   | 50    | 2              | UPAccess, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator, UPWorklist                          |
| UPDocGeneration            | API  | POST /up/doc_generation             | active   | 24    | 8              | UPAccess, UPDataUtil, UPDocService, UPErrorHandler, UPSecurity, UPValidator                        |
| UPFeedFile                 | API  | POST /up/generate-feedfile          | active   | 178   | 9              | UPDataUtil, UPErrorHandler, UPSecurity, UPShaper, UPValidator                                      |
| UPGetPlans                 | API  | POST /up/get-plans                  | active   | 114   | 8              | UPAccess, UPDataUtil, UPErrorHandler, UPProductData, UPSecurity, UPShaper, UPValidator             |
| UPIssue                    | API  | POST /up/issue                      | active   | 58    | 10             | UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPSecurity, UPShaper, UPValidator                  |
| UPListMasterTable          | API  | POST /up/list-master-table          | active   | 37    | 7              | UPDataUtil, UPErrorHandler, UPSecurity, UPValidator                                                |
| UPLoadPolicy               | API  | GET /up/load-policy                 | active   | 37    | 8              | UPAccess, UPDataUtil, UPErrorHandler, UPShaper, UPValidator                                        |
| UPMe                       | API  | GET /up/me                          | active   | 13    | 2              | UPAccess, UPErrorHandler                                                                           |
| UPProposal                 | API  | POST /up/proposal                   | active   | 44    | 7              | UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPProductData, UPSecurity, UPShaper, UPValidator   |
| UPProposalReject           | API  | POST /up/proposal-reject            | active   | 50    | 10             | UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPSecurity, UPValidator                            |
| UPShare                    | API  | POST /up/share                      | active   | 72    | 2              | UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPProductData, UPSecurity, UPValidator, UPWorklist |
| UPSpikeWho                 | API  | POST /up/spike-who                  | inactive | 20    | 6              |                                                                                                    |

| Name             | Kind     | Route                    | Status | Lines | Server version | Calls (common functions)                                                                                 |
|------------------|----------|--------------------------|--------|-------|----------------|----------------------------------------------------------------------------------------------------------|
| UPUpdateProposal | API      | POST /up/update-proposal | active | 62    | 10             | UPAccess, UPDataUtil, UPErrorHandler, UPProductData, UPSecurity, UPShaper, UPValidator                   |
| UPWorklistApi    | API      | POST /up/worklist        | active | 35    | 2              | UPAccess, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator, UPWorklist                                |
| UPAccess         | Function | -                        | active | 124   | 4              | UPDataUtil                                                                                               |
| UPCancelService  | Function | -                        | active | 364   | 8              | UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPPlatformClient, UPProductData, UPValidator, UPWorklist |
| UPDataUtil       | Function | -                        | active | 235   | 9              |                                                                                                          |
| UPDocBuilder     | Function | -                        | active | 195   | 3              | UPDataUtil, UPProductData                                                                                |
| UPDocService     | Function | -                        | active | 105   | 4              | UPDataUtil, UPDocBuilder, UPErrorHandler, UPShaper, UPValidator                                          |
| UPErrorHandler   | Function | -                        | active | 112   | 10             | UPDataUtil                                                                                               |
| UPNotify         | Function | -                        | active | 166   | 4              | UPAccess, UPDataUtil, UPProductData                                                                      |
| UPPlatformClient | Function | -                        | active | 49    | 7              | UPDataUtil                                                                                               |
| UPProductData    | Function | -                        | active | 213   | 5              | UPDataUtil, UPSecurity, UPShaper                                                                         |
| UPSecurity       | Function | -                        | active | 116   | 5              | UPDataUtil                                                                                               |
| UPShaper         | Function | -                        | active | 126   | 3              | UPDataUtil, UPProductData                                                                                |
| UPValidator      | Function | -                        | active | 334   | 3              | UPDataUtil, UPProductData                                                                                |
| UPWorklist       | Function | -                        | active | 408   | 4              | UPAccess, UPDataUtil, UPPlatformClient, UPProductData, UPValidator                                       |

Total: 3587 lines of Groovy (1040 in APIs, 2547 in functions).

## 4.2 Common Functions

### UPAccess (124 lines)

UPAccess - who is calling and what may they do (Maker / Proposal Checker / Cancellation Checker). The caller is taken from the platform token (IComposerAppContext), never from the request. Roles come from the UP\_UserRole table. Enforcement is switched by UP\_ApiConfig RoleEnforcement (TrueValue = on); with it off the APIs behave as before.

tables read: UP\_UserRole | config keys used: AllowSelfApproval, MsgForbidden, MsgNotOwner, MsgSelfApproval, RoleCancellationChecker, RoleEnforcement, RoleMaker, RoleProposalChecker | calls: UPDataUtil

| Method      | Parameters | Returns             | Purpose                                                                   |
|-------------|------------|---------------------|---------------------------------------------------------------------------|
| currentUser |            | Map<String, Object> | The platform user behind the token: UserName, UserId, DisplayName, Email. |

| Method            | Parameters                                       | Returns                   | Purpose                                                                                                                                               |
|-------------------|--------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| roleRows          | String userName, String userId                   | List<Map<String, Object>> | Active role rows of a user (by user name, case-insensitive, or by platform user id).                                                                  |
| me                |                                                  | Map<String, Object>       | Caller identity with roles, branch and contact details. Roles is empty when the user is not in UP_UserRole.                                           |
| require           | List<String> allowedRoles                        | Map<String, Object>       | Stops the call (HTTP 403, UP-403) unless the caller holds one of the roles. Returns the caller identity. With RoleEnforcement off nothing is checked. |
| requireNotCreator | Map<String, Object> caller, String creatorUserId | void                      | Maker-checker rule: the person who created an item cannot approve or reject it (unless AllowSelfApproval is on).                                      |
| requireAny        |                                                  | Map<String, Object>       | Any role holder (read access to policy data).                                                                                                         |
| requireCreator    | Map<String, Object> caller, String creatorUserId | void                      | A maker may only change what they created.                                                                                                            |
| holdersOf         | String role                                      | List<Map<String, Object>> | Role holders (for staff alerts): rows with the role that have an email or mobile number.                                                              |
| contactOf         | String userId                                    | Map<String, Object>       | One user's contact details by platform user id (the maker of a proposal); empty map when unknown.                                                     |

### UPCancelService (364 lines)

UPCancelService - cancellation in two steps (Checker, then Approver). \* The request itself is the platform's own pending cancellation endorsement: the Checker creates it (the endorsement number is the RequestNo), the Approver approves it (calculate, validate, issue) or rejects it (suspend). No premium is calculated by the platform here; the refund is calculated by UPDataUtil.refundCalc from the product rules. Every code (statuses, types, paths) comes from UP\_ApiConfig.

tables read: ProductMaster | SDK: ProposalSdkClient | config keys used: AllowSelfApproval, CancellInstallmentFlag, CancellInstallmentType, CancelPayMode, CancelReasonCode, CancelTypeFreeLook, CancelTypeProRata, DecisionApprove, EndoBasePath, EndoTypeCancel, PlatformErrorMaxLength, PolicyStatusInforce, ProposalLoadPath, RefundTypeFreeLook, RefundTypeProRata, RequestStatusApproved, RequestStatusPending, RequestStatusRejected, RoleEnforcement | calls: UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPPlatformClient, UPProductData, UPValidator, UPWorklist

| Method              | Parameters                      | Returns             | Purpose                                                                                                                         |
|---------------------|---------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| resolvePolicy       | Map<String, Object> input       | Map<String, Object> | The issued policy named by PolicyNo or ProposalNo, checked against the caller's channel. Returns null when it does not exist.   |
| pendingCancellation | Map<String, Object> policy      | Map<String, Object> | The pending cancellation endorsement of the policy, or null. Another kind of pending endorsement blocks the request.            |
| cancellationDate    | Map<String, Object> endorsement | LocalDate           | Cancellation date used for the refund: the endorsement effective date, or the start of the policy for a free-look cancellation. |
| policyInfo          | Map<String, Object> policy      | Map<String, Object> | Policy information block shared by both responses.                                                                              |
| policyStatus        | Map<String, Object> policy      | Map<String, Object> |                                                                                                                                 |

| Method        | Parameters                                                                        | Returns             | Purpose                                                                                                                                                                                                                                                                                                                            |
|---------------|-----------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| createRequest | Map<String, Object> policy, String effectiveDate, boolean freeLook, String reason | Map<String, Object> | Creates the pending cancellation endorsement (the request). No premium is calculated by the platform here.                                                                                                                                                                                                                         |
| suspend       | Map<String, Object> endorsement                                                   | void                | Withdraws a pending endorsement (reject, or before it is replaced on approval).                                                                                                                                                                                                                                                    |
| check         | Map<String, Object> input                                                         | Map<String, Object> | Checker: creates (or returns) the pending cancellation request and shows premium and refund details.                                                                                                                                                                                                                               |
| detail        | Map<String, Object> input                                                         | Map<String, Object> | Checker view of a pending request (nothing is created): who asked, why, and the refund an approval would pay.                                                                                                                                                                                                                      |
| decide        | Map<String, Object> input                                                         | Map<String, Object> | Approver: approves (cancels the policy through the platform) or rejects the pending request named by RequestNo.                                                                                                                                                                                                                    |
| refundCalc    | Map<String, Object> policy, java.time.LocalDate cancelDate                        | Map<String, Object> | Refund rule for a cancellation: full premium within the free-look period (counted from the commencement date), otherwise the percentage configured in UP_ProductRule (ProRataRefundPercent) of the pro-rata premium. Premium = plan premium on the risk (before tax). FreeLookDays and ProRataRefundPercent come from ProductRule. |

### UPDataUtil (235 lines)

UPDataUtil - core helpers: configuration (UP\_ApiConfig), table access, flag values, formats, error helpers.

tables read: UP\_ApiConfig | SDK: TablesSdkClient | config keys used: AllowedHosts, DateFormatDoc, FalseValue, MoneyFormat, MoneyScale, NumberFormat, PercentBase, RoundingMode, TrueValue

| Method             | Parameters                              | Returns                | Purpose                                                                                                                             |
|--------------------|-----------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| fail               | String message                          | void                   |                                                                                                                                     |
| str                | Object value                            | String                 |                                                                                                                                     |
| yes                |                                         | String                 | Flag values come from UP_ApiConfig (TrueValue / FalseValue), never from literals in code.                                           |
| no                 |                                         | String                 |                                                                                                                                     |
| isYes              | Object value                            | boolean                |                                                                                                                                     |
| isNo               | Object value                            | boolean                |                                                                                                                                     |
| requireAllowedHost | String url                              | void                   | The URL's host must be listed in UP_ApiConfig AllowedHosts and the scheme must be https before a bearer token is sent to it.        |
| fill               | String text, Map<String, String> values | String                 | Replaces {name} markers in a configured text.                                                                                       |
| roundingMode       |                                         | java.math.RoundingMode |                                                                                                                                     |
| moneyScale         |                                         | int                    |                                                                                                                                     |
| percentBase        |                                         | BigDecimal             |                                                                                                                                     |
| configMap          | String scope                            | Map<String, String>    | Configuration value from UP_ApiConfig (scope = product code or ""). A missing key is a configuration error, never a silent default. |

| Method          | Parameters                                    | Returns                   | Purpose                                                                                                                                            |
|-----------------|-----------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| cfgValue        | Map<String, String> config, String key        | String                    |                                                                                                                                                    |
| cfg             | String key, String scope                      | String                    |                                                                                                                                                    |
| pathValue       | Object root, String path                      | Object                    | Value at a path such as PolicyLobList[0].PolicyRiskList[0].DateOfBirth.                                                                            |
| fieldError      | String field, String code, String message     | Map<String, Object>       |                                                                                                                                                    |
| throwValidation | List<Map<String, Object>> errors              | void                      |                                                                                                                                                    |
| parseDate       | Object value                                  | LocalDate                 |                                                                                                                                                    |
| getRecords      | String tableName                              | List<Map<String, Object>> | All records of a datatable as plain maps. Read once per request (kept in the request attributes), so data changes are visible on the next request. |
| readRecords     | String tableName                              | List<Map<String, Object>> |                                                                                                                                                    |
| filterRecords   | String tableName, Map<String, Object> filters | List<Map<String, Object>> |                                                                                                                                                    |
| toDecimal       | Object value                                  | BigDecimal                |                                                                                                                                                    |
| fmtDate         | Object value                                  | String                    | Display date, pattern from UP_ApiConfig DateFormatDoc.                                                                                             |
| fmtMoney        | Object value                                  | String                    | Display amount, pattern from UP_ApiConfig MoneyFormat.                                                                                             |
| fmtNumber       | String v                                      | String                    | Display number, pattern from UP_ApiConfig NumberFormat.                                                                                            |

### UPDocBuilder (195 lines)

UPDocBuilder - print data for the application form / certificate, document wording, commission view.

tables read: UP\_BenefitText, UP\_DocWording | config keys used: PlanLabelPrefix | calls: UPDataUtil, UPProductData

| Method             | Parameters                                   | Returns             | Purpose                                                                                                                             |
|--------------------|----------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| describe           | String tableName, Object id                  | String              |                                                                                                                                     |
| benefitText        | Map<String, Object> benefit                  | String              | Benefit limit wording from UP_DocWording (ModeCode = BASIS: <LimitBasis>, falling back to the BASIS:Default row of the same table). |
| docBenefit         | String productCode, String code              | List<String>        | Official English / Arabic benefit names (UP_BenefitText): [english, arabic]. A row is required for every benefit that is printed.   |
| premiumWording     | String modeCode, String term, String premium | Map<String, Object> | English / Arabic wording of the premium frequency, premium text and period of cover for a premium mode (UP_DocWording).             |
| buildPrintData     | Map<String, Object> policy, String docType   | Map<String, Object> |                                                                                                                                     |
| rate               | Object commission, Object premium            | String              |                                                                                                                                     |
| commissionOfPolicy | Map<String, Object> policy                   | Map<String, Object> | Commission view of one loaded policy (commission is read from the stored policy, not from configuration).                           |

## UPDocService (105 lines)

UPDocService - builds the application form or the policy certificate as a PDF through the print service. Shared by UPDocGeneration and UPApplicationDocGeneration. Every setting (template version, print URL, export type, storage, mode, file name) comes from UP\_ApiConfig, the template name from UP\_DocTemplate.

tables read: DocTemplate | SDK: ProposalSdkClient | config keys used: DocContentType, DocFileNamePattern, DocTypeApplication, DocTypePolicy, PrintExportType, PrintMode, PrintStorage, PrintUrl, TemplateVersion | calls: UPDataUtil, UPDocBuilder, UPErrorHandler, UPShaper, UPValidator

| Method   | Parameters                                                | Returns | Purpose                                                                                        |
|----------|-----------------------------------------------------------|---------|------------------------------------------------------------------------------------------------|
| generate | String apiName, String docType, Map<String, Object> input | Object  | Returns the PDF (HttpEntity), the print data when PreviewData is the TrueValue, or a 404 body. |

## UPErrorHandler (112 lines)

UPErrorHandler - turns an exception into a structured error response and sets the HTTP status. Validation problems return 400, missing role or maker-checker violations 403, state conflicts 409, platform API errors keep their status; configuration problems and anything else are 500 with a generic table-driven message (the detail goes to the log, the caller gets the trace id).

config keys used: MsgConfigError, MsgInvalidNumber, MsgPlatformInput, MsgServerError | calls: UPDataUtil

| Method         | Parameters                     | Returns             | Purpose                                                                                                                                           |
|----------------|--------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| configText     | String key, String technical   | String              | Message text from UP_ApiConfig; the technical fallback is used only when the configuration itself cannot be read.                                 |
| isInternalText | String message                 | boolean             | True when a platform message carries database or framework internals (index names, class names, parser positions) that must not reach the caller. |
| handle         | String apiName, Exception ex   | Map<String, Object> |                                                                                                                                                   |
| notFound       | String apiName, String message | Map<String, Object> |                                                                                                                                                   |

## UPNotify (166 lines)

UPNotify - Email / SMS through the tenant's SNS service, driven by UP\_NotifyTemplate. A notification never fails the business call: every attempt returns a result row {Event, Recipient, Channel, To (masked), Status}. Status: SENT, DRYRUN (NotifyMode is not LIVE), NO\_ADDRESS (no email / mobile known), NO\_SMS\_TEMPLATE (SmsTemplateCode is NONE) or FAILED. Email is sent with subject and body from the template. The SNS SMS request carries no free text: it names an SMS template (SmsTemplateCode) of the SNS account and its parameters.

tables read: UP\_NotifyTemplate | SDK: SnsSdkClient | config keys used: EmailAccountName, NotifyMode, PortalUrl, SmsAccountName | calls: UPAccess, UPDataUtil, UPProductData

| Method     | Parameters                                                                                             | Returns             | Purpose                                                                           |
|------------|--------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------|
| maskEmail  | String email                                                                                           | String              |                                                                                   |
| maskMobile | String mobile                                                                                          | String              |                                                                                   |
| render     | String text, Map<String, String> params                                                                | String              | Template text with {Name} markers replaced; markers without a value become empty. |
| resultRow  | String event, String recipient, String channel, String to, String status                               | Map<String, Object> |                                                                                   |
| sendOne    | String channel, String to, String subject, String body, String smsTemplate, Map<String, String> params | String              |                                                                                   |

| Method     | Parameters                                                                                                     | Returns                   | Purpose                                                                                                                                                                                                                                                      |
|------------|----------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fire       | String event, Map<String, String> params, Map<String, Object> customer, String makerUserId, String onlyChannel | List<Map<String, Object>> | Sends every active template of the event. Recipients: a role (all active holders in UP_UserRole), MAKER (the creator of the item, by platform user id) or CUSTOMER (customer map with Email / Mobile). onlyChannels restricts to EMAIL / SMS (empty = both). |
| paramsOf   | Map<String, Object> policy, Map<String, String> extra                                                          | Map<String, String>       | Placeholder values for the templates from a policy / proposal map (stored or shaped). extra overrides / adds values.                                                                                                                                         |
| customerOf | Map<String, Object> storedPolicy                                                                               | Map<String, Object>       | Customer contact (Email, Mobile) of a stored proposal / policy, unmasked, for customer alerts.                                                                                                                                                               |

### UPPlatformClient (49 lines)

UPPlatformClient - calls platform services (endorsement) with the caller's own token. Returns { status: <http status>, body: <response text> } and never throws for HTTP or platform errors.

config keys used: GatewayUrl | calls: UPDataUtil

| Method | Parameters                              | Returns             | Purpose |
|--------|-----------------------------------------|---------------------|---------|
| call   | String method, String path, Object body | Map<String, Object> |         |
| parse  | Map<String, Object> response            | Map<String, Object> |         |

### UPProductData (213 lines)

UPProductData - product master data (variants, rules, plans, benefits) and preparation of the proposal payload.

tables read: ProductMaster, ProductPlanRelation, ProductRule, UP\_Carrier, UP\_PlanBenefitRelation | config keys used: RiskElementCode | calls: UPDataUtil, UPSecurity, UPShaper

| Method         | Parameters                                                     | Returns                   | Purpose                                                                                                                                                                                                                                                                        |
|----------------|----------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| getVariant     | String productCode, String subcode                             | Map<String, Object>       |                                                                                                                                                                                                                                                                                |
| carrierOf      | String productCode, String subcode                             | Map<String, Object>       | Insurer (carrier) of a product variant: the code is on the product (UP_ProductMaster.CarrierCode), the name comes from UP_Carrier. A product without a carrier, or a carrier that is missing or inactive in UP_Carrier, is a configuration error (UP-CONFIG), never a default. |
| addCarrier     | Map<String, Object> target, String productCode, String subcode | void                      | Adds CarrierCode and CarrierName (the insurer of the product) to a response map.                                                                                                                                                                                               |
| getRule        | String productCode, String subcode                             | Map<String, Object>       |                                                                                                                                                                                                                                                                                |
| allowedModes   | Map<String, Object> rule                                       | List<String>              |                                                                                                                                                                                                                                                                                |
| getPlanRows    | String productCode, String subcode, String premiumModeCode     | List<Map<String, Object>> | Plan rows (premium + commission) for the variant and premium mode.                                                                                                                                                                                                             |
| getBenefitRows | String productCode, String subcode, String planCode            | List<Map<String, Object>> |                                                                                                                                                                                                                                                                                |
| buildPlanView  | String productCode, String subcode, String premiumModeCode     | List<Map<String, Object>> | Plans for the front end: premium and benefits only, never commission.                                                                                                                                                                                                          |

| Method          | Parameters                         | Returns             | Purpose                                                                                                                                                                                            |
|-----------------|------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prepareProposal | Map<String, Object> input          | Map<String, Object> | Validates the request against product data and fills the authoritative values (plan premium, commission, coverages when none are supplied). Returns the payload ready for the native proposal SDK. |
| variantName     | String productCode, String subcode | String              | Display name of a product variant (UP_ProductMaster CarrierProductName); empty when unknown.                                                                                                       |

### UPSecurity (116 lines)

UPSecurity - request hygiene shared by every API. readInput the body must be a JSON object (anything else is a 400, never a 500); it is scanned before any API logic runs scan blocked characters in every text value, extra allow-list for free-text fields (400 INVALID\_CHARACTER) stripServerOwned fields the server owns (policy number, ids, status) are never taken from the caller defaultOrgCode a proposal always carries a branch that the platform code table PubBranch accepts The baselines below are security controls: when UP\_ApiConfig has the key (BlockedCharacters, FreeTextFields, FreeTextPattern, ServerOwnedFields, OrgCode) the table value is used, otherwise the built-in baseline applies (never an error).

*calls: UPDataUtil*

| Method                 | Parameters                                                                                                                                          | Returns             | Purpose                                                                                                                                                                            |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| blockedBaseline        |                                                                                                                                                     | String              | Control characters and the characters used for script, template, shell and expression injection.                                                                                   |
| freeTextBaseline       |                                                                                                                                                     | String              | Free-text fields: letters of any script, digits, space and . , ' - / @ + ( ) only.                                                                                                 |
| freeTextFieldsBaseline |                                                                                                                                                     | String              |                                                                                                                                                                                    |
| serverOwnedBaseline    |                                                                                                                                                     | String              |                                                                                                                                                                                    |
| orgCodeBaseline        |                                                                                                                                                     | String              |                                                                                                                                                                                    |
| maxFindings            |                                                                                                                                                     | int                 |                                                                                                                                                                                    |
| setting                | String key, String baseline                                                                                                                         | String              | Optional table override of a baseline.                                                                                                                                             |
| readInput              | Object body                                                                                                                                         | Map<String, Object> | The request body as a map, already checked by scan(). Null is an empty request; an array, text or number is a 400.                                                                 |
| scan                   | Map<String, Object> input                                                                                                                           | void                | Rejects blocked characters in any text value, and anything outside the allow-list in the free-text fields. The value is never echoed.                                              |
| scanValue              | Object value, String path, String key, Pattern blocked, Pattern freeText, Set<String> freeFields, List<Map<String, Object>> errors, UPDataUtil util | void                |                                                                                                                                                                                    |
| stripServerOwned       | Map<String, Object> input                                                                                                                           | void                | Removes the top-level fields the server owns. The platform sets the proposal status itself; a status sent by the caller made the platform fail (HTTP 500 / 502), so it is dropped. |
| defaultOrgCode         | Map<String, Object> input                                                                                                                           | void                | A branch the platform accepts (code table PubBranch); the caller's own OrgCode is kept.                                                                                            |

### UPShaper (126 lines)

UPShaper - shapes the responses returned to callers (commission, premium fields, payment info).

config keys used: InstalmentAmount, InstalmentFeeSeq, InstalmentPayRate, InstalmentPeriodCount, InstalmentPeriodSeq, PolicyStatusEffective, ReturnCommission, SequenceNumberStart, ZeroPremiumFields, ZeroPremiumValue | calls: UPDataUtil, UPProductData

| Method              | Parameters              | Returns | Purpose                                                                                                                                |
|---------------------|-------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| stripCommission     | Object node             | Object  | Removes commission from a policy / proposal body before it is returned to a front-end caller.                                          |
| restorePayment      | Object body             | Object  | Rebuilds PolicyPaymentInfoList from the flattened payment fields.                                                                      |
| returnCommission    |                         | boolean | Commission is returned only while UP_ApiConfig ReturnCommission equals the TrueValue.                                                  |
| enrichPremiumFields | Map<String, Object> map | void    | Adds the premium structure fields the PA001 responses carry (zero where the level is not rated) and the plan row premium / commission. |
| shapeResponse       | Object body             | Object  | Response shaping for every front-end facing API: same structure as the PA001 flow, payment info in its original shape.                 |

### UPValidator (334 lines)

UPValidator - request validation: field rules (UP\_FieldRule), cross-field and product rules, state / channel checks.

tables read: ProductMaster, UP\_FieldRule | config keys used: BeneficiaryShareTotal, EffectiveDateMinOffsetDays, MasterTables, NonSingleTermYears, PatternMaxInput, PatternMaxLength, ProposalStatusIssued, ProposalStatusRejected, SinglePremiumModeCode | calls: UPDataUtil, UPProductData

| Method              | Parameters                                                                                                    | Returns                   | Purpose                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requireOpenProposal | Map<String, Object> stored, String action                                                                     | void                      | A proposal can only be updated / issued / rejected while it is open (status codes come from UP_ApiConfig).                                                                        |
| requireProductMatch | Map<String, Object> stored, String productCode, String subcode                                                | void                      | The stored proposal / policy must belong to the product the caller names.                                                                                                         |
| requireChannel      | Map<String, Object> stored, String channelCode                                                                | void                      | The stored proposal / policy must have been created by the channel that calls (ChannelCode is saved on every proposal).                                                           |
| codeExists          | String table, String column, String value                                                                     | boolean                   |                                                                                                                                                                                   |
| validate            | String api, Map<String, Object> input                                                                         | void                      | Field level validation. Rules come from UP_FieldRule (per API) and product rules from UP_ProductRule. All problems are collected and returned together as HTTP 400 with errors[]. |
| crossFieldErrors    | String api, Map<String, Object> input, boolean productFieldBad                                                | List<Map<String, Object>> | Rules that need several fields or product data. Only runs on values that already passed the field rules.                                                                          |
| entryErrors         | Map<String, Object> rule, String dateOfBirth, String effectiveDate, String premiumModeCode, String policyTerm | List<Map<String, Object>> | Age, term and premium mode checks from UP_ProductRule, returned as field errors.                                                                                                  |

### UPWorklist (408 lines)

UPWorklist - queues and dashboard figures read from the platform's search index (nothing is stored by this API). Proposal queue : proposals by ProposalStatus (pending / issued / rejected), per UP product. Cancellation queue : cancellation endorsements by EndoStatus (pending / issued). Maker filter : the maker's platform user id is stamped on every proposal in AgentCode by UPProductData.prepareProposal, so "my submissions" is an index

search. The index is limited to 1000 rows per query (page size x page number) and holds no premium, so premium and commission are read from the stored policies (capped by DashboardMaxPolicies).

tables read: ProductMaster | SDK: ProposalSdkClient | config keys used: DashboardMaxPolicies, DayEndTime, EndoSearchModule, EndoSearchPath, EndoStatusIssued, EndoStatusPending, EndoTypeCancelFilter, PolicyStatusInforce, ProposalStatusIssued, ProposalStatusPending, ProposalStatusRejected, SearchModule, SearchSortField, SearchSortType, WorklistMaxRows, WorklistPageSize | calls: UPAccess, UPDataUtil, UPPlatformClient, UPProductData, UPValidator

| Method             | Parameters                                                                                                                | Returns             | Purpose                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| productCodes       |                                                                                                                           | List<String>        | Distinct product codes enabled for the APIs (UP_ProductMaster).                                                                                                             |
| proposalStatusCode | String name                                                                                                               | String              | Status text for a status filter name, as configured. Returns "" for ALL.                                                                                                    |
| dateRange          | Map<String, Object> input, String key, boolean from                                                                       | Map<String, Object> |                                                                                                                                                                             |
| condition          | Map<String, Object> conditions, Map<String, Object> from, Map<String, Object> to, int pageNo, int pageSize, String module | SearchCondition     |                                                                                                                                                                             |
| policySearch       | Map<String, Object> conditions, Map<String, Object> from, Map<String, Object> to, int pageNo, int pageSize                | Map<String, Object> | One policy-index search: {Total, Docs}.                                                                                                                                     |
| endoSearch         | Map<String, Object> conditions, Map<String, Object> from, Map<String, Object> to, int pageNo, int pageSize                | Map<String, Object> | One endorsement-index search: {Total, Docs}.                                                                                                                                |
| loadPolicy         | String proposalNo, String policyNo                                                                                        | Map<String, Object> |                                                                                                                                                                             |
| makerName          | String userId                                                                                                             | String              | Display name of a platform user id (UP_UserRole), else the id itself.                                                                                                       |
| policyRow          | Map<String, Object> policy                                                                                                | Map<String, Object> | Row shown in a queue, from the stored policy / proposal.                                                                                                                    |
| intOf              | Object value, int fallback                                                                                                | int                 |                                                                                                                                                                             |
| proposalList       | Map<String, Object> input, String mineUserId                                                                              | Map<String, Object> | Proposal queue: type PROPOSAL (all makers) or MINE (mineUserId is the caller's id), filtered by Status, product and dates.                                                  |
| cancellationList   | Map<String, Object> input                                                                                                 | Map<String, Object> | Cancellation queue: pending (default) or issued cancellation endorsements.                                                                                                  |
| summary            | Map<String, Object> input, String mineUserId                                                                              | Map<String, Object> | Counts of proposals and cancellations in the date window (mineUserId limits proposals to one maker).                                                                        |
| commissionSection  | Map<String, Object> input, String productCode, String mineUserId                                                          | Map<String, Object> | Premium and commission of the issued policies of one product in the window, by plan. Capped at DashboardMaxPolicies newest policies (Truncated tells when the cap applied). |

## 4.3 APIs

### UPApplicationDocGeneration — POST /up/application\_doc\_generation (25 lines)

Generate the application form of a proposal as a PDF (same request as the PA001 flow). Request: { "proposalNo": "...", "ProductSubcode": "1" } The template is looked up in UP\_DocTemplate by ProductCode + ProductSubcode + DocType.

calls: UPAccess, UPDataUtil, UPDocService, UPErrorHandler, UPSecurity, UPValidator | config keys: DocTypeApplication

### UPCancelApprove — POST /up/cancel-approve (25 lines)

Cancellation - Approver. Approves (cancels the policy) or rejects the pending request created by the Checker. Request: { "ChannelCode": "SOHAR", "PolicyNo" | "ProposalNo": "...", "RequestNo": "<from the Checker>", "Decision": "APPROVE" | "REJECT" } Response: RequestNo, RequestStatus, PolicyInfo, PolicyStatus, RefundPremium.

*calls: UPAccess, UPCancelService, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator | config keys: RoleCancellationChecker*

### UPCancelCheck — POST /up/cancel-check (26 lines)

Cancellation - Checker. Creates the pending cancellation request for an issued policy and shows what an approval would do. Request: { "ChannelCode": "SOHAR", "PolicyNo" | "ProposalNo": "...", "CancellationDate": "2026-10-20", "CancelReason": "..." } Response: RequestNo, RequestStatus, PolicyInfo, PolicyStatus (current), PremiumDetails, RefundDetails. Repeating the call for a policy that already has a pending request returns that request (RequestExists = TrueValue).

*calls: UPAccess, UPCancelService, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator | config keys: RoleMaker*

### UPCancelDetail — POST /up/cancel-detail (28 lines)

Cancellation - details of a pending request for the Cancellation Checker (and the maker who raised it). Creates and changes nothing. Request: { "ChannelCode": "SOHAR", "PolicyNo" | "ProposalNo": "..." } Response: RequestNo, RequestedBy, RequestedOn, CancelReason, CancellationDate, PolicyInfo, PremiumDetails, RefundDetails, CanDecide.

*calls: UPAccess, UPCancelService, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator | config keys: RoleCancellationChecker, RoleMaker*

### UPCancelPolicy — POST /up/cancel-policy (5 lines)

Retired (inactive). Cancellation is now done in two steps: UPCancelCheck (/up/cancel-check) and UPCancelApprove (/up/cancel-approve).

### UPCommissionQuery — POST /up/commission-query (137 lines)

Commission query - the only API that returns commission (all other APIs hide it). \* 1. By policy: { "PolicyNo": "POFP00100000013" } 2. By configuration: { "ProductCode": "FP001", "ProductSubcode": "1", "PlanId": "2" (optional) } 3. By period: { "ProductCode": "FP001" (optional), "FromDate": "2026-10-01", "ToDate": "2026-10-31", "PageNo": 1, "PageSize": 50 } \* Commission is configured per product / variant / plan in the ProductPlanRelation table (column Commission). It is written onto the policy when the premium is calculated, so a policy keeps the commission it was issued with.

*calls: UPAccess, UPDataUtil, UPDocBuilder, UPErrorHandler, UPProductData, UPSecurity, UPValidator | SDK: ProposalSdkClient | tables: ProductPlanRelation | config keys: CommissionDefaultPageNo, CommissionDefaultPageSize, DayEndTime, DayStartTime, ProposalStatusIssued, SearchModule, SearchSortField, SearchSortType*

### UPDashboard — POST /up/dashboard (50 lines)

Dashboard figures. Two sections so the page can load them in parallel: SUMMARY counts of proposals (submitted / pending / issued / rejected, pending older than a day) and cancellations, per product. COMMISSION premium and commission of the issued policies of ONE product (ProductCode) in the window, by plan; cancelled policies are shown apart. Request: { "ChannelCode", "Section", "ProductCode" (COMMISSION), "Scope": "ALL" | "MINE", "FromDate", "ToDate" } (default window: the last 30 days, at most DashboardMaxDays). A caller who is only a Maker always sees their own submissions (Scope MINE).

*calls: UPAccess, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator, UPWorklist | config keys: DashboardMaxDays, RoleMaker*

### UPDocGeneration — POST /up/doc\_generation (24 lines)

Generate the application form or the policy certificate as a PDF. Request: { "DocType": "Application" | "Policy", "ProposalNo": "...", "PolicyNo": "..." }

*calls: UPAccess, UPDataUtil, UPDocService, UPErrorHandler, UPSecurity, UPValidator*

### **UPFeedFile — POST /up/generate-feedfile (178 lines)**

Generate the policy feed file (Excel by default, JSON when OutputFormat = "JSON"). Request: { "Conditions": {"ProductCode": "FP001"}, "FromRangeConditions": {"EffectiveDate": "..."}, "ToRangeConditions": {"EffectiveDate": "..."}, "SearchType": "Policy", "PageNo": 1, "PageSize": 50, "OutputFormat": "XLSX" | "JSON" }  
Commission is not part of the feed file.

*calls: UPDataUtil, UPErrorHandler, UPSecurity, UPShaper, UPValidator | SDK: ProposalSdkClient | tables: UP\_FeedColumn | config keys: FeedContentType, FeedFileName, FeedMaxPages, FeedMaxRecords, FeedSheetName, ProposalStatusEntry, ProposalStatusIssued, SearchModule, SearchSortField, SearchSortType*

### **UPGetPlans — POST /up/get-plans (114 lines)**

Get plans for a product variant. Same shape as the PA001 flow: the request is the policy body (ProductCode + ProductSubcode, insured details), the response is that policy with every plan of the variant in PolicyRiskList[].PlanList (PlanId, PlanName, DuePremium, Commission, PolicyCoverageList). Optional: PremiumModeCode, PolicyTerm. Age and term rules are checked when the date of birth is supplied.

*calls: UPAccess, UPDataUtil, UPErrorHandler, UPProductData, UPSecurity, UPShaper, UPValidator | SDK: QuotationSdkClient | config keys: BusinessCateCode, CustomerFlagNo, GetPlansPlatformValidate, IsAutoAttachedForm, IsPremiumCalcSuccess, PolicyStatusEffective, RiskElementCode, SequenceNumberStart, ZeroPremiumFields, ZeroPremiumValue*

### **UPIssue — POST /up/issue (58 lines)**

Issue a policy from a proposal. Request: { "ProposalNo": "...", "ProductCode": "..." }.

*calls: UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPSecurity, UPShaper, UPValidator | SDK: ProposalSdkClient | config keys: RoleProposalChecker*

### **UPListMasterTable — POST /up/list-master-table (37 lines)**

List master data for the front-end dropdowns. Request: { "DataTableName": "Relationship", "Id": "2" (optional) }.

*calls: UPDataUtil, UPErrorHandler, UPSecurity, UPValidator | config keys: MasterTables*

### **UPLoadPolicy — GET /up/load-policy (37 lines)**

Load a policy (policyNo) or a proposal (proposalNo). Commission is removed from the response.

*calls: UPAccess, UPDataUtil, UPErrorHandler, UPShaper, UPValidator | SDK: ProposalSdkClient*

### **UPMe — GET /up/me (13 lines)**

Who am I: the caller taken from the platform token with the roles configured in UP\_UserRole. Response: UserName, UserId, DisplayName, Email, Branch, Roles[] (MAKER, PROPOSAL\_CHECKER, CANCELLATION\_CHECKER), RoleEnforcement, Insurer.

*calls: UPAccess, UPErrorHandler*

### **UPProposal — POST /up/proposal (44 lines)**

Create a proposal (application). The request uses the native policy JSON (PolicyLobList / PolicyRiskList / PlanList / PolicyCoverageList / BeneficiaryList ...). The plan premium is taken from product data; coverages are filled from the plan when not supplied.

*calls: UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPProductData, UPSecurity, UPShaper, UPValidator | SDK: ProposalSdkClient | config keys: RoleMaker*

### **UPProposalReject — POST /up/proposal-reject (50 lines)**

Reject a proposal (checker action). Request: { "ProposalNo", "ProposalRejectDesc" }.

*calls: UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPSecurity, UPValidator | SDK: ProposalSdkClient | config keys: RoleProposalChecker, SuccessStatusText*

### **UPShare — POST /up/share (72 lines)**

Share a quotation, proposal or policy with the customer by Email and / or SMS (InsureMO SNS; dry run until the tenant has an account, see NotifyMode). Request: { "ChannelCode", "DocType": "QUOTATION" | "PROPOSAL" | "POLICY", "Channel": "EMAIL" | "SMS" | "BOTH", "Email", "Mobile", "ProposalNo" | "PolicyNo" (PROPOSAL / POLICY), "Details": { InsuredName, ProductName, PlanName, Premium, EffectiveDate } (QUOTATION) } Without Email / Mobile the customer details stored on the proposal / policy are used. Response: DocType, Channel, Notifications[] ({Event, Recipient, Channel, To (masked), Status}).

*calls: UPAccess, UPDataUtil, UPErrorHandler, UPNotify, UPProductData, UPSecurity, UPValidator, UPWorklist | config keys: ShareDocQuotation*

### **UPSpikeWho — POST /up/spike-who (20 lines)**

Inactive probe API from the identity and SNS spike; delete it before UAT (section 19.4).

### **UPUpdateProposal — POST /up/update-proposal (62 lines)**

Update an existing proposal. ProposalNo is required; the policy identifiers are loaded from the stored proposal.

*calls: UPAccess, UPDataUtil, UPErrorHandler, UPProductData, UPSecurity, UPShaper, UPValidator | SDK: ProposalSdkClient | config keys: RoleMaker*

### **UPWorklistApi — POST /up/worklist (35 lines)**

Work queues for bank staff, read from the platform search index. Request: { "ChannelCode": "SOHAR", "Type": "PROPOSAL" | "CANCELLATION" | "MINE", "Status": "PENDING" | "ISSUED" | "REJECTED" | "ALL" (cancellations: PENDING | ISSUED), "ProductCode", "FromDate", "ToDate", "PageNo", "PageSize" } PROPOSAL = the Proposal Checker's queue, CANCELLATION = the Cancellation Checker's queue (Maker may read it), MINE = the caller's own submissions (Maker). Response: Type, Status, PageNo, PageSize, Total, Records[].

*calls: UPAccess, UPDataUtil, UPErrorHandler, UPSecurity, UPValidator, UPWorklist | config keys: RoleCancellationChecker, RoleMaker, RoleProposalChecker*

## 5. API Walkthroughs

What each API does in code, in the order it does it. Names in backticks are functions or methods listed in section 4.

| API                           | Flow in code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1.2 changes to existing APIs | UPProposal: UPAccess.require(Maker), prepareProposal, then AgentCode = caller id, create, shape, UPNotify.fire("SUBMITTED"), Notifications[] added. UPUdateProposal: Maker and creator only (requireCreator), keeps the stored AgentCode. UPIssue: Proposal Checker, requireNotCreator(existing.AgentCode), after the issue PROPOSAL_ISSUED to the maker and the customer (contact from the raw loaded proposal). UPProposalReject: Proposal Checker, not the creator, ProposalRejectDesc is required by a field rule, PROPOSAL_REJECTED to the maker. UPCancelCheck: Maker; a new request sends CANCEL_REQUESTED. UPCancelApprove: Cancellation Checker; UPCancelService.decide compares the requester (DataEntryUserId of the pending endorsement) with the caller and sends CANCEL_APPROVED / CANCEL_REJECTED. UPGetPlans, UPLoadPolicy, UPDocGeneration, UPApplicationDocGeneration and UPCommissionQuery call requireAny().                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| UPMe                          | UPAccess.me(); no validation, no role needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| UPWorklistApi                 | Validates (UP_FieldRule rows for api name UPWorklist), role by Type, then UPWorklist.proposalList (PROPOSAL, MINE) or cancellationList. Section 15.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| UPDashboard                   | Validates, requireAny(), forces own figures for a maker-only caller, checks the window, then UPWorklist.summary or commissionSection. Section 15.5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| UPShare                       | Validates, requireAny(), builds the placeholders (quotation: from Details; proposal / policy: loaded and channel-checked), takes the contact from the request or the stored record and calls UPNotify.fire(event, ..., onlyChannel). Section 16.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| UPCancelDetail                | Validates, Maker or Cancellation Checker, UPCancelService.detail: resolves the policy, finds the pending endorsement (signed policy id), recalculates the refund, returns requester, reason and CanDecide; creates nothing; 404 when no request is pending.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| UPGetPlans                    | Validates the body (validator.validate("UPGetPlans"): product, variant, premium mode, term, age from DateOfBirth). Reads the variant (UPProductData.getVariant) and rule (getRule) and builds the PlanList directly from the tables: getPlanRows (UP_ProductPlanRelation: plan id, name, premium, commission) and getBenefitRows (UP_PlanBenefitRelation: code, description, sum insured) — no platform call. The premium shown here is therefore the table premium; the proposal premium is calculated by the platform rating <CODE>_PREM_CALC from the same tables, so the two stay equal as long as the data is consistent. Defaults of the policy skeleton (status, zero premium fields, flags, customer flags) come from UP_ApiConfig (PolicyStatusEffective, ZeroPremiumFields, CustomerFlagNo, ...). Like PA001, it first runs the platform's own quotation validation (QuotationSdkClient.quotationApi().newValidateRequestBuilder(), empty event code and language, a copy of the request) when UP_ApiConfig.GetPlansPlatformValidate = Y (MC: Y); the result is not used, a rejection stops the request (HTTP 400, code UP-PLATFORM-VALIDATION for policy validation errors such as a code value outside its code table, or the platform's own 400). The plans themselves are still built without a platform call. It does not use shapeResponse, so CarrierCode / CarrierName are added with products.addCarrier before the return. |
| UPProposal                    | Rejects ProposalNo (409-style 400), validates, then UPProductData.prepareProposal: drops client PlanList and DuePremium (the platform rating builds plan, covers, premium and commission), removes server-owned fields and defaults OrgCode (UPSecurity), flattens PolicyPaymentInfoList into Payment* policy fields (the platform instalment rating would otherwise re-round the premium) and returns the payload. The API then calls proposalApi().newApplicationRequestBuilder() and returns shaper.shapeResponse, which rebuilds the payment structure, applies the commission switch and adds the insurer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| UPUpdateProposal              | Same preparation as create. Loads the stored proposal by ProposalNo (404), checks channel, product match and that it is open (409), copies PolicyId, PolicyElementId and VersionSeq of the stored record into the payload, calls newSaveRequestBuilder() and then newPersistCalculateRequestBuilder() because the save drops the calculated plan row.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| UPIssue                       | Loads the proposal (404), checks channel, product match (MISMATCH) and open state (409). When the optional request field Recalculate = Y it first runs persistCalculate on the stored proposal (not part of the public specification). Then newIssuePolicyRequestBuilder issues it and the shaped policy with the new PolicyNo is returned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| UPLoadPolicy                  | GET. Exactly one of policyNo / proposalNo (400); loads with code descriptions (withCodeDesc), 404 when missing, requireChannel, shapeResponse.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| API                                          | Flow in code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UPProposalReject                             | Loads the proposal (404), checks channel and open state (409), calls <code>newRejectRequestBuilder</code> with a <code>RejectVo</code> (proposal number and reason) and returns <code>{status: SUCCESS (SuccessStatusText), message: &lt;the reason&gt;}</code> .                                                                                                                                                                                                                                                                                  |
| UPDocGeneration / UPApplicationDocGeneration | Thin wrappers over <code>UPDocService.generate</code> : resolve the template name from <code>UP_DocTemplate</code> (product, sub-product, DocType), build the print data with <code>UPDocBuilder</code> (wording from <code>UP_DocWording</code> , benefit names from <code>UP_BenefitText</code> , formats from <code>UP_ApiConfig</code> ), call the print service at <code>PrintUrl</code> (host checked against <code>AllowedHosts</code> ) with the caller's token and return the PDF (or the print data when <code>PreviewData = Y</code> ). |
| UPCancelCheck / UPCancelApprove              | See section 8 for the complete sequence: <code>UPCancelService.check</code> and <code>decide</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| UPListMasterTable                            | Allows only tables listed in <code>UP_ApiConfig.MasterTables</code> ; returns all records or the one with <code>Id</code> (commission column removed).                                                                                                                                                                                                                                                                                                                                                                                             |
| UPCommissionQuery                            | Three modes: by <code>PolicyNo</code> ( <code>UPDocBuilder.commissionOfPolicy</code> ), by product configuration (rows of <code>UP_ProductPlanRelation</code> with premium, commission and rate), or for a period (search on the platform, load each policy, keep only the caller's channel, page totals). Page size is capped by <code>CommissionMaxPageSize</code> .                                                                                                                                                                             |
| UPFeedFile                                   | Searches policies with the platform query API (period, product, paging), loads each hit, keeps the caller's channel, shapes it and writes the columns defined in <code>UP_FeedColumn</code> (heading + source path, alternatives separated by <code> </code> ) to an Excel sheet or to JSON. Capped by <code>FeedMaxRecords / FeedMaxPages</code> (400 <code>RANGE_T00_LARGE</code> ).                                                                                                                                                             |

## 6. Data and Configuration Reference

### 6.1 Tables

| Table                  | Table id (MC) | Rows | Columns                                                                                                                                                                                                                                                | Purpose                                                                                                                                                                                            | Read by                                                  |
|------------------------|---------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| UP_ApiConfig           | 1188575596    | 113  | Api, Scope, Key, Value                                                                                                                                                                                                                                 | Key / value configuration of the code (Api, Scope, Key, Value).<br>configMap(scope) merges * rows with product-specific rows.                                                                      | UPDataUtil.cfg /<br>cfgValue, every<br>function          |
| UP_FieldRule           | 1188575333    | 167  | Api, Path, Label, Required, DataType, MinLength, MaxLength, Pattern, MinValue, MaxValue, CodeTable, CodeColumn, EnumValues, AppliesTo, WhenPath, WhenValue                                                                                             | Validation rules per API and field path: Required, DataType, MinLength, MaxLength, Pattern, MinValue, MaxValue, EnumValues, CodeTable + CodeColumn, WhenPath + WhenValue (conditional), AppliesTo. | UPValidator.validate                                     |
| UP_ProductMaster       | 1188475672    | 11   | Description, LOB, IMOProductCode, IsRenewable, IsActive, IsFlatRate, Premium, SumInsured, HasPlans, CarrierCode, CarrierProductCode, CarrierProductName, IsResident, BenefitRelationTableName, LobId, CarrierName                                      | Product variants (IMOProductCode, CarrierProductCode = sub-product, name, LOB, CarrierCode, IsActive).                                                                                             | UPProductData.getVariant, carrierOf                      |
| UP_ProductPlanRelation | 1188475676    | 57   | IMOProductCode, IMOPlanCode, IMOPlanName, CarrierCode, CarrierProductCode, CarrierProductName, CarrierPlanCode, CarrierPlanName, IsActive, Premium, SumInsured, Commission, PremiumModeCode                                                            | Plans per variant with premium, commission, premium mode.                                                                                                                                          | UPProductData, UPGetPlans, commission configuration view |
| UP_PlanBenefitRelation | 1188474890    | 148  | IMOProductCode, IMOPlanCode, CarrierCode, CarrierProductCode, CarrierPlanCode, BenefitCode, BenefitDesc, BenefitLevel, ParentBenefitCode, BenefitLimit, BenefitLimitDesc, IsMainBenefit, IsAdditionalBenefit, Notes, IsApplicable, LimitBasis, MaxDays | Covers and sums insured per plan.                                                                                                                                                                  | UPProductData, UPGetPlans, documents                     |

| Table             | Table id (MC) | Rows | Columns                                                                                                                                                                         | Purpose                                                                                                                                                                                                        | Read by                                             |
|-------------------|---------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| UP_Product Rule   | 1188475684    | 11   | ProductCode, ProductSubcode, PremiumModes, MinTerm, MaxTerm, MinAge, MaxAge, MaxEntryAge, MaxExitAge, MandatoryDocuments, SinglePremiumRule, FreeLookDays, ProRataRefundPercent | Per variant: MinAge, MaxAge, MaxEntryAge, MaxExitAge, MinTerm, MaxTerm, PremiumModes, FreeLookDays, ProRataRefundPercent, mandatory documents.                                                                 | UPValidator.entryErrors, UPCancelService.refundCalc |
| UP_DocTemplate    | 1188475689    | 22   | DocumentCode, TemplateName, FileName, ProductCode, ProductSubcode, DocType                                                                                                      | Output template name per product, sub-product and document type.                                                                                                                                               | UPDocService                                        |
| UP_DocWording     | 1188556760    | 27   | ModeCode, Key, Value                                                                                                                                                            | Wording for premium modes and benefit limit bases (BASIS: *), English / Arabic.                                                                                                                                | UPDocBuilder                                        |
| UP_BenefitText    | 1188575620    | 21   | ProductCode, BenefitCode, NameEn, NameAr                                                                                                                                        | Official English / Arabic benefit names per product.                                                                                                                                                           | UPDocBuilder                                        |
| UP_FeedColumn     | 1188591810    | 29   | Sequence, Heading, Source                                                                                                                                                       | Columns of the policy feed file (Sequence, Heading, Source).                                                                                                                                                   | UPFeedFile                                          |
| UP_Channel        | 1188591925    | 1    | ChannelCode, Description, IsActive                                                                                                                                              | Registered channels (ChannelCode, Description, IsActive).                                                                                                                                                      | UPValidator (ChannelCode rule)                      |
| UP_Carrier        | 1188663015    | 1    | CarrierCode, CarrierName, IsActive                                                                                                                                              | Insurers (CarrierCode, CarrierName, IsActive); linked from UP_ProductMaster.CarrierCode.                                                                                                                       | UPProductData.carrierOf                             |
| UP_UserRole       | 1188710773    | 3    | UserRoleKey, UserName, UserId, Role, DisplayName, Branch, Email, Mobile, IsActive                                                                                               | Staff roles: UserRoleKey (user ROLE), UserName, UserId (platform id), Role (MAKER / PROPOSAL_CHECKER / CANCELLATION_CHECKER), DisplayName, Branch, Email, Mobile, IsActive. Never exposed through master data. | UPAccess                                            |
| UP_NotifyTemplate | 1188710781    | 22   | TemplateKey, Event, Recipient, Channel, Subject, Body, SmsTemplateCode, IsActive                                                                                                | Email / SMS texts per event, recipient and channel (TemplateKey, Event, Recipient, Channel, Subject, Body, SmsTemplateCode, IsActive).                                                                         | UPNotify                                            |

The code maps a few logical names to the UP\_ tables (UPDataUtil.readRecords): ProductMaster, ProductPlanRelation, UP\_PlanBenefitRelation, ProductRule and DocTemplate. Tables are read once per request and cached in the request attributes, so a changed row is visible on the next request without any deployment.

## 6.2 Configuration Keys (UP\_ApiConfig)

Every key with its current MC value and the functions that read it. Environment specific keys are marked in section 13.

| Key                   | Value (MC)             | Used in                                     |
|-----------------------|------------------------|---------------------------------------------|
| AllowSelfApproval     | Y                      | UPAccess, UPCancelService                   |
| AllowedHosts          | portal-gw.insuremo.com | UPDataUtil                                  |
| BeneficiaryShareTotal | 100                    | UPValidator                                 |
| BusinessCateCode      | 1                      | UPGetPlans                                  |
| CancelFreeLookLabel   | Free look              | (read through a pattern or documented only) |

| Key                        | Value (MC)                                                        | Used in                                     |
|----------------------------|-------------------------------------------------------------------|---------------------------------------------|
| CancelInstallmentFlag      | Y                                                                 | UPCancelService                             |
| CancelInstallmentType      | 1                                                                 | UPCancelService                             |
| CancelPayMode              | 30                                                                | UPCancelService                             |
| CancelReasonCode           | 1                                                                 | UPCancelService                             |
| CancelTypeFreeLook         | 31                                                                | UPCancelService                             |
| CancelTypeProRata          | 32                                                                | UPCancelService                             |
| CommissionDefaultPageNo    | 1                                                                 | UPCommissionQuery                           |
| CommissionDefaultPageSize  | 50                                                                | UPCommissionQuery                           |
| CommissionMaxPageSize      | 50                                                                | (read through a pattern or documented only) |
| CustomerFlagNo             | N                                                                 | UPGetPlans                                  |
| DashboardMaxDays           | 180                                                               | UPDashboard                                 |
| DashboardMaxPolicies       | 60                                                                | UPWorklist                                  |
| DateFormatDoc              | dd/MM/yyyy                                                        | UPDataUtil                                  |
| DayEndTime                 | T23:59:59                                                         | UPCommissionQuery, UPWorklist               |
| DayStartTime               | T00:00:00                                                         | UPCommissionQuery, UPWorklist               |
| DecisionApprove            | APPROVE                                                           | UPCancelService                             |
| DecisionReject             | REJECT                                                            | (read through a pattern or documented only) |
| DocContentType             | application/pdf                                                   | UPDocService                                |
| DocFileNamePattern         | {docType}_{number}.pdf                                            | UPDocService                                |
| DocTypeApplication         | Application                                                       | UPApplicationDocGeneration, UPDocService    |
| DocTypePolicy              | Policy                                                            | UPDocService                                |
| EffectiveDateMinOffsetDays | 0                                                                 | UPValidator                                 |
| EmailAccountName           | NONE                                                              | UPNotify                                    |
| EndoBasePath               | /platform/endo/core/endo/v1                                       | UPCancelService                             |
| EndoSearchModule           | Endorsement                                                       | UPWorklist                                  |
| EndoSearchPath             | /platform/endo/v1/query                                           | UPWorklist                                  |
| EndoStatusIssued           | 300                                                               | UPWorklist                                  |
| EndoStatusPending          | 120                                                               | UPWorklist                                  |
| EndoStatusWithdrawn        | 600                                                               | (read through a pattern or documented only) |
| EndoTypeCancel             | 3                                                                 | UPCancelService                             |
| EndoTypeCancelFilter       | 3                                                                 | UPWorklist                                  |
| FalseValue                 | N                                                                 | UPDataUtil                                  |
| FeedContentType            | application/vnd.openxmlformats-officedocument.spreadsheetml.sheet | UPFeedFile                                  |
| FeedFileMaxPageSize        | 50                                                                | (read through a pattern or documented only) |
| FeedFileName               | Policy_Feed.xlsx                                                  | UPFeedFile                                  |
| FeedMaxPages               | 40                                                                | UPFeedFile                                  |
| FeedMaxRecords             | 2000                                                              | UPFeedFile                                  |
| FeedSheetName              | PolicyFeed                                                        | UPFeedFile                                  |
| GatewayUrl                 | https://portal-gw.insuremo.com                                    | UPPlatformClient                            |
| InstalmentAmount           | 0                                                                 | UPShaper                                    |
| InstalmentFeeSeq           | 1                                                                 | UPShaper                                    |

| Key                    | Value (MC)                                                                                                                                            | Used in                                                |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| InstalmentPayRate      | 1                                                                                                                                                     | UPShaper                                               |
| InstalmentPeriodCount  | 1                                                                                                                                                     | UPShaper                                               |
| InstalmentPeriodSeq    | 1                                                                                                                                                     | UPShaper                                               |
| IsAutoAttachedForm     | Y                                                                                                                                                     | UPGetPlans                                             |
| IsPremiumCalcSuccess   | Y                                                                                                                                                     | UPGetPlans                                             |
| MasterTables           | BranchCode, CustomerType, GenderCode, Id Type, InsMaritalStatus, Plan, Pre...                                                                         | UPListMasterTable, UPValidator                         |
| MoneyFormat            | #,##0.00                                                                                                                                              | UPDataUtil                                             |
| MoneyScale             | 2                                                                                                                                                     | UPDataUtil                                             |
| MsgConfigError         | The service is not configured correctly. Please contact support and...                                                                                | UPErrorHandler                                         |
| MsgForbidden           | You do not have the role required for this action.                                                                                                    | UPAccess                                               |
| MsgInvalidNumber       | A numeric value in the request is not valid.                                                                                                          | UPErrorHandler                                         |
| MsgNotOwner            | Only the maker who created this proposal can change it.                                                                                               | UPAccess                                               |
| MsgNotifyDryRun        | Notification recorded (dry run, nothing was sent).                                                                                                    | (read through a pattern or documented only)            |
| MsgSelfApproval        | The person who created an item cannot approve or reject it.                                                                                           | UPAccess                                               |
| MsgServerError         | The request could not be processed. Please contact support and quot...                                                                                | UPErrorHandler                                         |
| NonSingleTermYears     | 1                                                                                                                                                     | UPValidator                                            |
| NotifyMode             | DRYRUN                                                                                                                                                | UPNotify                                               |
| NumberFormat           | #,##0.##                                                                                                                                              | UPDataUtil                                             |
| PatternMaxInput        | 500                                                                                                                                                   | UPValidator                                            |
| PatternMaxLength       | 200                                                                                                                                                   | UPValidator                                            |
| PercentBase            | 100                                                                                                                                                   | UPDataUtil                                             |
| PlanLabelPrefix        | PLAN-                                                                                                                                                 | UPDocBuilder                                           |
| PlatformErrorMaxLength | 300                                                                                                                                                   | UPCancelService                                        |
| PolicyStatusEffective  | 2                                                                                                                                                     | UPGetPlans, UPShaper                                   |
| PolicyStatusInforce    | 2                                                                                                                                                     | UPCancelService, UPWorklist                            |
| PortalUrl              | <a href="https://portal.insuremo.com/uic-web/#!/p/sohar-insurance-portal">https://portal.insuremo.com/uic-web/#!/p/sohar-insurance-portal</a>         | UPNotify                                               |
| PrintExportType        | pdf                                                                                                                                                   | UPDocService                                           |
| PrintMode              | sync                                                                                                                                                  | UPDocService                                           |
| PrintStorage           | not_store                                                                                                                                             | UPDocService                                           |
| PrintUrl               | <a href="https://portal-gw.insuremo.com/mofo/1.0/print/v3/printtask/generate">https://portal-gw.insuremo.com/mofo/1.0/print/v3/printtask/generate</a> | UPDocService                                           |
| ProposalLoadPath       | /platform/proposal/v1/load                                                                                                                            | UPCancelService                                        |
| ProposalStatusEntry    | 1                                                                                                                                                     | UPFeedFile                                             |
| ProposalStatusIssued   | 3                                                                                                                                                     | UPCommissionQuery, UPFeedFile, UPValidator, UPWorklist |
| ProposalStatusIssued   | 3                                                                                                                                                     | UPCommissionQuery, UPFeedFile, UPValidator, UPWorklist |
| ProposalStatusPending  | 2                                                                                                                                                     | UPWorklist                                             |
| ProposalStatusRejected | 4                                                                                                                                                     | UPValidator, UPWorklist                                |

| Key                     | Value (MC)                                                                 | Used in                                                                                           |
|-------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| ProposalStatusRejected  | 4                                                                          | UPValidator, UPWorklist                                                                           |
| RefundTypeFreeLook      | FREE_LOOK                                                                  | UPCancelService                                                                                   |
| RefundTypeProRata       | PRO_RATA                                                                   | UPCancelService                                                                                   |
| RequestStatusApproved   | APPROVED                                                                   | UPCancelService                                                                                   |
| RequestStatusPending    | PENDING                                                                    | UPCancelService                                                                                   |
| RequestStatusRejected   | REJECTED                                                                   | UPCancelService                                                                                   |
| ReturnCommission        | Y                                                                          | UPShaper                                                                                          |
| RiskElementCode         | R00001                                                                     | UPGetPlans, UPProductData                                                                         |
| RoleCancellationChecker | CANCELLATION_CHECKER                                                       | UPAccess, UPCancelApprove, UPCancelDetail, UPWorklistApi                                          |
| RoleEnforcement         | Y                                                                          | UPAccess, UPCancelService                                                                         |
| RoleMaker               | MAKER                                                                      | UPAccess, UPCancelCheck, UPCancelDetail, UPDashboard, UPProposal, UPUpdateProposal, UPWorklistApi |
| RoleProposalChecker     | PROPOSAL_CHECKER                                                           | UPAccess, UPIssue, UPProposalReject, UPWorklistApi                                                |
| RoundingMode            | HALF_UP                                                                    | UPDataUtil                                                                                        |
| SearchModule            | Policy                                                                     | UPCommissionQuery, UPFeedFile, UPWorklist                                                         |
| SearchPath              | /platform/proposal/v1/queryPolicy                                          | (read through a pattern or documented only)                                                       |
| SearchSortField         | index_time                                                                 | UPCommissionQuery, UPFeedFile, UPWorklist                                                         |
| SearchSortType          | desc                                                                       | UPCommissionQuery, UPFeedFile, UPWorklist                                                         |
| SequenceNumberStart     | 1                                                                          | UPGetPlans, UPShaper                                                                              |
| ShareDocPolicy          | SHARE_POLICY                                                               | UPShare                                                                                           |
| ShareDocProposal        | SHARE_PROPOSAL                                                             | UPShare                                                                                           |
| ShareDocQuotation       | SHARE_QUOTATION                                                            | UPShare                                                                                           |
| ShareMaxDetailLength    | 80                                                                         | (read through a pattern or documented only)                                                       |
| SinglePremiumModeCode   | 1                                                                          | UPValidator                                                                                       |
| SmsAccountName          | NONE                                                                       | UPNotify                                                                                          |
| SuccessStatusText       | SUCCESS                                                                    | UPProposalReject                                                                                  |
| TemplateVersion         | 1                                                                          | UPDocService                                                                                      |
| TrueValue               | Y                                                                          | UPDataUtil                                                                                        |
| WorklistMaxRows         | 1000                                                                       | UPWorklist                                                                                        |
| WorklistPageSize        | 20                                                                         | UPWorklist                                                                                        |
| ZeroPremiumFields       | AdjustedPremium,AnnualPremium,BeforeVa<br>tPremium,GrossPremium,Standar... | UPGetPlans, UPShaper                                                                              |
| ZeroPremiumValue        | 0                                                                          | UPGetPlans, UPShaper                                                                              |

### 6.3 Field Rules (UP\_FieldRule)

| API                        | Rules |
|----------------------------|-------|
| UPApplicationDocGeneration | 3     |
| UPCancelApprove            | 5     |
| UPCancelCheck              | 5     |
| UPCancelDetail             | 3     |
| UPCommissionQuery          | 9     |
| UPDashboard                | 6     |

| API               | Rules |
|-------------------|-------|
| UPDocGeneration   | 6     |
| UPFeedFile        | 8     |
| UPGetPlans        | 8     |
| UPIssue           | 3     |
| UPListMasterTable | 3     |
| UPLoadPolicy      | 3     |
| UPProposal        | 40    |
| UPProposalReject  | 4     |
| UPShare           | 12    |
| UPUpdateProposal  | 41    |
| UPWorklist        | 8     |

A rule row is evaluated by `UPValidator.validate`: Path uses indexes (`PolicyLobList[0].PolicyRiskList[0].InsuredName`); Required Y/N; `DataType` string / integer / decimal / date / enum; `Pattern` is a regular expression (length and input size are capped by `PatternMaxLength` / `PatternMaxInput`); `CodeTable` + `CodeColumn` check the value against a table (`UP_Channel`, `UP_Carrier`, `BranchCode`, ...); `WhenPath` + `WhenValue` make the rule conditional (for example `PolicyNo` is required when `DocType = Policy`); `AppliesTo` limits a rule to a product. Changing a rule is a row change and needs no deployment.

## 6.4 Security Baselines (UPSecurity)

| Control                                     | Baseline in code                                                                                     | Optional override key in UP_ApiConfig |
|---------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------|
| Blocked characters in any text value        | control characters and <code>&lt; &gt; \$ { } \   ^ ~ ; `</code>                                     | BlockedCharacters                     |
| Free-text fields                            | InsuredName, Address, BeneficiaryName, CancelReason, OtherCancelReason, ProposalRejectDesc           | FreeTextFields                        |
| Free-text allow-list                        | letters of any script, digits, space and <code>. , ' - / @ + ( )</code>                              | FreeTextPattern                       |
| Server-owned fields removed from a proposal | PolicyNo, PolicyId, PolicyElementId, ProposalStatus, PolicyStatus, AgentCode (v1.2: the maker stamp) | ServerOwnedFields                     |
| Default OrgCode                             | 1                                                                                                    | OrgCode                               |

*These are the only values that have a built-in default; a missing override key never fails a request. Moving them fully into tables (and treating a missing key as UP-CONFIG) is on the backlog.*

## 7. Platform Configuration (Products)

### 7.1 What Exists on the Platform

For each of the seven product codes the platform holds a tech product `<CODE>_TECH` and a market product `<CODE>` (ids in the deployment plan), the policy data dictionary tree (Policy, PolicyLob, PolicyRisk R00001, Plan, PolicyCoverage, Beneficiary), a premium rating definition `<CODE>_PREM_CALC` bound to the market product, the plan and cover data in the `UP_` tables, and the output templates named in `UP_DocTemplate` (22 rows).

### 7.2 Data Dictionary Bindings

The new products bind only the fields the flows need (24 policy-level, 17 tech-level, 8 risk-level, 4 plan, 6 coverage, 4 beneficiary fields) at product level (RecordUsage 5). PA001 binds far more (408 policy-level fields). The cancellation endorsement rating additionally reads delta fields; `fix_endo_fields.py` bound the ones the platform reported as missing (`StandardGrossPremiumDelta`, `AnnualPremiumDelta`, `ShortRate` on the risk and coverage objects) on the new tech products only. The script never creates `VAT` (there is no VAT in this solution) and refuses to touch any context that is not one of the seven tech products.

### 7.3 How the Products Were Built

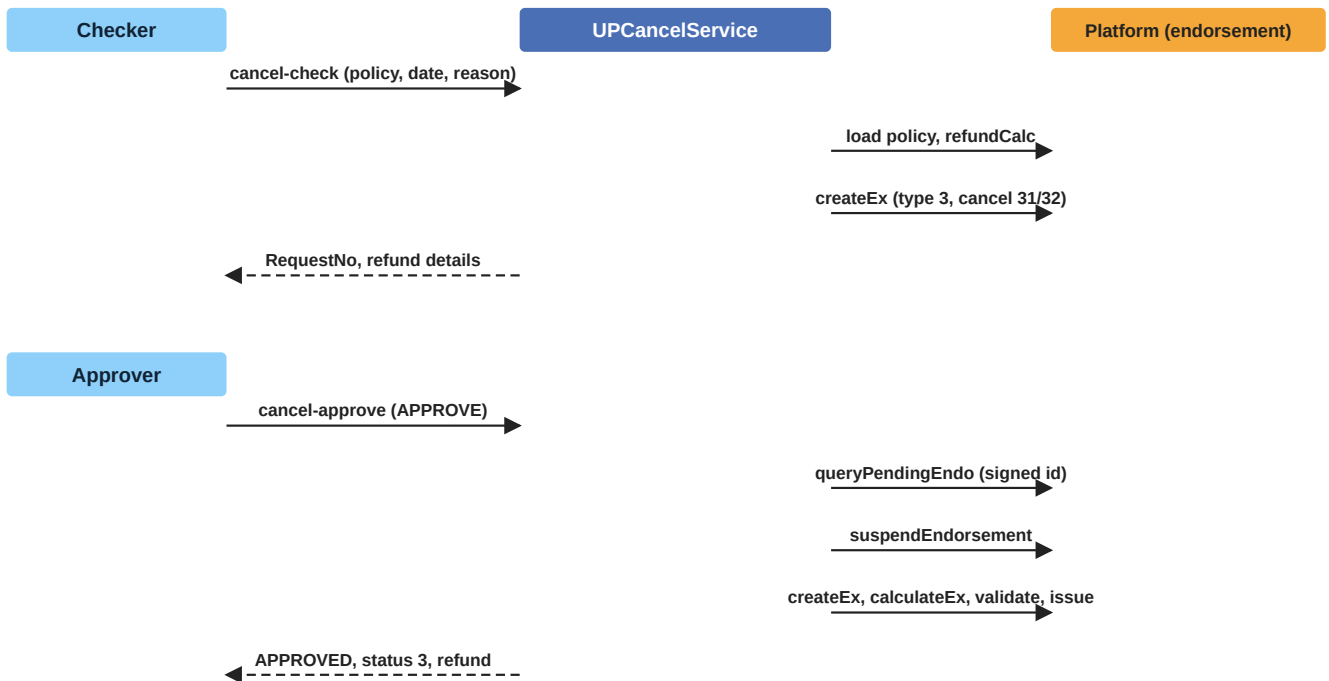
| Step                                                   | Tool                                                                                                                                                                                            |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tech and market products, tree, DD fields, code tables | <code>build_product.py &lt;spec.json&gt; &lt;step&gt;</code> with <code>spec_&lt;CODE&gt;.json</code>                                                                                           |
| Rating definition per product                          | <code>rating_build.py &lt;CODE&gt; --confirm</code> (copy of <code>PA001_PREM_CALC</code> bound to the market product), patched by <code>patch_rating.py</code> / <code>patch_rating2.py</code> |
| Tables and rows                                        | <code>make_tables.py</code> , <code>data_rows.py</code> , <code>uptab.py</code> (create table, save rows), <code>make_rules.py</code> (field rules), <code>stage_*_rows.py</code> (config keys) |
| Output templates                                       | <code>make_templates2.py</code> builds the bilingual HTML templates from the form designs; <code>gen_all_tpl.py</code> / <code>os_gen.py</code> upload and test them                            |
| Cancellation fields                                    | <code>fix_endo_fields.py</code> , <code>bind_cancel_field.py</code> (optional <code>IsAdminChange</code> , not used)                                                                            |
| PA001 guard                                            | <code>snapshot.py before after</code>                                                                                                                                                           |

## 8. Business Algorithms

### 8.1 Premium Mode, Term and Age

UPValidator.entryErrors reads the variant's UP\_ProductRule row: the premium mode must be in PremiumModes; single premium (SinglePremiumModeCode) needs a term between MinTerm and MaxTerm, other modes need NonSingleTermYears (1); the age at the effective date must be within MinAge / MaxAge and MaxEntryAge, and age + term within MaxExitAge. PremiumModeCode is mandatory where the plans of a variant differ by mode (UPPA001, TL001, CI001, LP001). Premium and commission are never taken from the request: the platform rating calculates them from the plan tables.

### 8.2 Cancellation (Checker / Approver)



When the platform refuses a step the fresh endorsement stays pending: the answer is HTTP 409 and the request can be approved again or rejected.

Figure 2: Cancellation — Checker and Approver sequence in UPCancelService

The request is the platform's own pending cancellation endorsement; its number is the RequestNo. From v1.2 the Checker step is the **Maker** role and the Approver step the **Cancellation Checker** role (the code and routes are unchanged); the requester cannot decide their own request; cancel-detail shows a pending request without creating one; and the steps send alerts (section 16).

#### Checker — UPCancelService.check

1. resolvePolicy: by PolicyNo or ProposalNo (a proposal must be issued: 409), requireChannel, the product must be enabled (UP\_ProductMaster).
2. Policy must be in force (status PolicyStatusInforce); a cancelled policy is 409.
3. pendingCancellation: queries the pending endorsement with the **signed policy id** (id,signature) returned by the platform's own load (ProposalLoadPath), sent once-encoded. Another kind of pending endorsement blocks the request (409). An existing cancellation request is shown again with the date it was created with.
4. refundCalc(policy, date): free look when elapsed days <= FreeLookDays (refund 100%), otherwise premium x unexpired days / policy days x ProRataRefundPercent (rounded MoneyScale, RoundingMode).
5. createRequest: POST EndoBasePath/createEx with EndoType 3, CancelType 31 (free look) or 32 (pro-rata), reason, payment info; for free look the effective date is the policy start.
6. Response: RequestNo, RequestStatus PENDING, PolicyInfo (with insurer), PolicyStatus, PremiumDetails, RefundDetails.

## Approver — `UPCancelService.decide`

1. Same lookup; `RequestNo` must be the pending endorsement (404 otherwise, also on a second approval).
2. REJECT: `suspendEndorsement`, answer REJECTED.
3. APPROVE: suspend the saved (thin) request, create a **fresh** endorsement with the same terms (`createEx`), then `calculateEx`, `validate`, `issueEndorsement`. The saved request is replaced because `calculateEx` needs the full `NewPolicy`, which only `createEx` returns.
4. If the platform refuses a step the exception text is capped (`PlatformErrorMaxLength`) and returned as 409  
The platform refused the cancellation of `<policy>`; the request is still pending as `<EndoNo>`.
5. Success: status 3, `RefundPremium` (same calculation as the Checker), `EndorsementNo`, `RequestStatus` APPROVED.

## 8.3 Platform Quirks That Shaped the Code

| Quirk                                                                                                                                                                                | Effect / handling                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Endorsement service needs the signed policy id ( <code>id,signature</code> ) and rejects a double-encoded id                                                                         | Signed id taken from the platform load and sent once-encoded in <code>pendingCancellation</code> .                                                                                        |
| <code>calculateEx</code> needs the full <code>NewPolicy</code> ; a saved pending endorsement is thin                                                                                 | On APPROVE the saved request is suspended and a fresh <code>createEx</code> is used.                                                                                                      |
| The platform REST client throws its own business exception                                                                                                                           | <code>UPPlatformClient</code> returns <code>{status, body}</code> ; the approver answers 409 instead of 500.                                                                              |
| A policy created without <code>OrgCode</code> gets -1; <code>validate</code> of the endorsement checks <code>OrgCode</code> against code table <code>PubBranch</code> (only value 1) | <code>UPSecurity.defaultOrgCode</code> sets 1 when it is missing; callers send <code>OrgCode</code> 1.                                                                                    |
| A policy status sent by the caller made the platform fail (500 / 502); a caller policy number was accepted                                                                           | <code>UPSecurity.stripServerOwned</code> removes <code>PolicyNo</code> , <code>PolicyId</code> , <code>PolicyElementId</code> , <code>ProposalStatus</code> , <code>PolicyStatus</code> . |
| The cancellation rating reads delta fields that were not bound on the new products                                                                                                   | <code>fix_endo_fields.py</code> bound them on the seven tech products (section 7.2).                                                                                                      |
| The platform save drops the calculated plan row                                                                                                                                      | <code>UPUpdateProposal</code> calls <code>persistCalculate</code> after save.                                                                                                             |
| Platform instalment rating re-rounds the premium                                                                                                                                     | Payment info is flattened to <code>Payment*</code> fields and rebuilt in <code>UPShaper.restorePayment</code> .                                                                           |
| Responses contain a <code>TempData</code> block with masked / encrypted copies of ID and mobile                                                                                      | Not part of the contract; flagged for review (section 14).                                                                                                                                |
| Wrong HTTP method returns 422 from the gateway, not from the code                                                                                                                    | Documented in the specification.                                                                                                                                                          |

## 9. Errors, Logging and Tracing

| Situation                                                                                                                              | Exception thrown in code                                                                          | HTTP / code                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Validation problems                                                                                                                    | <code>IllegalArgumentException("VALIDATION:" + json)</code> via <code>util.throwValidation</code> | 400 UP-VALIDATION with <code>errors[]</code>                                                          |
| Simple bad request                                                                                                                     | <code>util.fail(msg) -&gt; IllegalArgumentException</code>                                        | 400 UP-400                                                                                            |
| State conflict                                                                                                                         | <code>IllegalArgumentException("CONFLICT:...")</code>                                             | 409 UP-409                                                                                            |
| Bad number                                                                                                                             | <code>NumberFormatException</code>                                                                | 400 UP-400, text <code>MsgInvalidNumber</code>                                                        |
| Missing configuration / table row                                                                                                      | <code>IllegalStateException</code>                                                                | 500 UP-CONFIG, text <code>MsgConfigError</code> ; detail in the log                                   |
| Platform business error                                                                                                                | <code>ApiException</code> (message sanitised by <code>isInternalText</code> )                     | status of the platform, its code; internal text replaced by <code>MsgServerError</code>               |
| Missing role / maker-checker breach                                                                                                    | <code>IllegalArgumentException("FORBIDDEN:...")</code> from <code>UPAccess</code>                 | 403 UP-403, texts <code>MsgForbidden</code> , <code>MsgSelfApproval</code> , <code>MsgNotOwner</code> |
| Platform policy validation rejected the request (a <code>MultipleBusinessException</code> whose text starts "Policy Validation Error") | handled in <code>UPErrorHandler.handle</code> by class name                                       | 400 UP-PLATFORM-VALIDATION, message = the platform text                                               |
| Not found                                                                                                                              | <code>errorHandler.notFound(api, msg)</code>                                                      | 404 UP-404                                                                                            |
| Anything else                                                                                                                          | <code>Exception</code>                                                                            | 500 UP-500, generic text; full exception logged with the trace id                                     |

Tracing: every error body carries `trace_id`. Find the call in the platform log with `imo log query` using that trace id (application logs of the iComposer runtime); UP-CONFIG messages name the missing key only in the log. Do not add `println` or secrets to the code; use the slf4j logger.

# 10. How-To Recipes

## Add or change a validation rule

1. Read the rows of `UP_FieldRule` (`imo config datatable record-list --id 1188575333`).
2. Add a row (`record-batch-save`, a new unique Id) or change one (`record-save` with its `RecordId`): `Api`, `Path`, `Label`, `Required`, `DataType`, `limits`, `Pattern`, `CodeTable`.
3. Test the API with a valid and an invalid value; no deployment is needed.
4. Update the specification (`api_spec_data.py` reads the rules) and the policy model.

## Rule that needs code (cross-field, product rule)

1. Edit `UPValidator.entryErrors / validate` (keep one `util.fieldError(path, code, message)` per problem).
2. Push `UPValidator`; push nothing else unless a signature changed.
3. Add a test case (`run_tests.py`, business cases) and a line in the specification.

## Change free look days or refund percent

1. Edit `FreeLookDays / ProRataRefundPercent` of the variant row in `UP_ProductRule` (`record-save`).
2. Check with `cancel-check` on a test policy; update the specification section 6.2 if the value differs from the documented 15 / 80.

## Add a plan or change a premium

1. Add or change rows in `UP_ProductPlanRelation` and `UP_PlanBenefitRelation`; add the plan id to the `Plan` code table if it is new.
2. Run `Get Plans` and `Create Proposal` for the plan and compare the premium with the table (the rating reads the tables).

## Add a product variant (sub-product)

1. Add `UP_ProductMaster`, `UP_ProductRule`, plan and cover rows, `UP_DocTemplate` rows and the output templates (`make_templates2.py`).
2. Add test samples (`samples/req/<TAG>_proposal.json`) so `ref_flow.py` can run the variant.

## Add a new product code

1. Create the tech and market product, the DD tree and bindings and the rating definition with `build_product.py` and `rating_build.py` (section 7.3).
2. Bind the cancellation delta fields with `fix_endo_fields.py <TAG>`.
3. Add the table rows as for a variant; add the code to the product lists of the tooling (`products.json`, test scripts).

## Add a channel

1. Add a row to `UP_Channel` (`ChannelCode`, `Description`, `IsActive Y`). The `ChannelCode` rule checks the value against this table.

## Add or change the insurer

1. Add a row to `UP_Carrier`; set `UP_ProductMaster.CarrierCode` of the variants to it. Responses show the name from the table; a product whose carrier row is missing or inactive returns `UP-CONFIG`.

## Add a master (code) table to Master Data

1. Append the table name to the value of `UP_ApiConfig.MasterTables` (`record-save` of the row with its `RecordId`).

## Add a field to a response

1. Policy-level: set it in `UPShaper.shapeResponse` (or the API for `Get Plans`, which does not use the shaper).  
Cancellation: `UPCancelService.policyInfo` and the `out` maps. Commission: `UPDocBuilder.commissionOfPolicy`. Feed: add a `UP_FeedColumn` row.
2. Add the field to the specification (`api_spec_data.py` reads recorded responses: re-run `ref_flow.py` for one plan per product).

## Add a new API

1. `imo icomposer create api ...` (path `/up/<name>`), copy the pattern of a small API (readInput, validate, call, shape, errorHandler).
2. Add `UP_FieldRule` rows for the new API name, push, add cases to `run_tests.py` and `security_probe.py`, document in the specification.

## Change a message text or format

1. Edit the `UP_ApiConfig` value (`MsgServerError`, `MsgConfigError`, `MsgInvalidNumber`, `DateFormatDoc`, `MoneyFormat`, ...).

## Move to another environment

1. Change `AllowedHosts`, `GatewayUrl`, `PrintUrl` (section 13) and re-check the ids listed in the deployment plan.

## Add a user or change a role

1. Section 19.2: rows in `UP_UserRole` (`record-batch-save` / `record-save`); no deployment.

## Change a message text or add an event

1. Edit the `Subject` / `Body` of the row in `UP_NotifyTemplate` (`record-save`). For a new event add rows (`TemplateKey = Event|Recipient|Channel`) and call `notify.fire("EVENT", notify.paramsOf(policy, extra), customer, makerUserId, "")` from the API; add the placeholders to `paramsOf` if needed.

## Send real email / SMS

1. Section 16.5.

## Add a menu item or screen to the portal

1. Add a `<section class="view" id="v_x">` and an entry in `NAV` (roles list or null for all); load its data in `go()`; call the API through `api()`; add a browser check to `flow2.js`; push with `imo uic page push`.

## Add a column to a queue

1. Add the field to `UPWorklist.policyRow`, push `UPWorklist`, then add a column object to `colsProposal` / `colsCancel` in the portal.

## Turn role enforcement on or off, or allow self approval

1. `python3 setcfg.py RoleEnforcement Y|N` and `AllowSelfApproval Y|N` (section 19.1).

## Apply the Sohar brand

1. Edit the variables in the first `:root` block of `index.html` (colours, font) and the text badge; push the page.

# 11. Test Tooling

All scripts live in `uponly-build` and read the token and gateway from `env.sh` (`source env.sh` sets `B`, `GW`, `TOK`; the token is fetched with `imo auth prepare --profile portal:uponly`). Never commit tokens.

| Script                          | What it does                                                                                                                                                                                                                                                             |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>carrier_check.py</code>   | One flow per product (first plan): <code>CarrierCode</code> / <code>CarrierName</code> in every API, wrong <code>CarrierCode</code> , master data and feed file; output <code>carrier/results.json</code> .                                                              |
| <code>ref_flow.py</code>        | Runs the 13-step flow (get plans ... cancel approve, load cancelled policy) for every plan of a variant; <code>ONLY_PLAN=1</code> limits it to one plan; output in <code>samples/ref/&lt;TAG&gt;_P&lt;plan&gt;/</code> .                                                 |
| <code>refresh_devdata.py</code> | Rebuilds <code>devdata/assets.json</code> and <code>devdata/tables.json</code> offline for this guide.                                                                                                                                                                   |
| <code>role_check.py</code>      | v1.2: 35 live checks of roles, maker-checker rule, queues, cancellation detail, notifications (dry run), share, dashboard; switches <code>UP_UserRole</code> rows and <code>AllowSelfApproval</code> during the run and restores them.                                   |
| <code>run_tests.py</code>       | Generates and runs the API test suite (1,085 cases: field validation, business rules, state transitions, cancellation incl. refund checks per product, master data, feed file, robustness). Results in <code>tests/results/</code> and <code>tests/results.json</code> . |
| <code>security_probe.py</code>  | Live security probe (token tampering, channel isolation, injection, tampering, resource abuse, headers, errors, request hygiene); output <code>security/results.json</code> .                                                                                            |
| <code>setcfg.py</code>          | Changes <code>UP_ApiConfig</code> values with <code>record-save</code> and prints the old value.                                                                                                                                                                         |
| <code>snapshot.py</code>        | Read-only snapshot of PA001 related objects; compare <code>before</code> and <code>after</code> to prove PA001 is unchanged.                                                                                                                                             |
| <code>workflow_setup.py</code>  | v1.2 set-up: creates <code>UP_UserRole</code> , <code>UP_NotifyTemplate</code> and the first configuration keys / field rules ( <code>workflow_setup2..5.py</code> add the later keys and rules).                                                                        |

Collections: `make_sohar_collection.py` (single Sohar Bank collection, one flow per product), `make_plan_collections.py` (one collection per plan with recorded failure examples), `make_cases_by_product.py`. Reports: `gen_report.py`, `make_excel.py`. Documents: `build_policy_model.py` (workbook), `spec_content.py` + `render_spec.py` / `render_docx.py` (specification), `dev_content12.py` + `dev12_sections.py` (this guide; the v1.1 generator is `dev_content.py`). Browser tests of the portal (`flow2.js`, `login.js`, `run.js`, Playwright) are described in section 17.6.

## All scripts in the build folder

| Script                             | Purpose (first docstring line)                                                                                                                                                                |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>api_spec_data.py</code>      | One structured specification per API (inputs with M / O / C, outputs with A / C), shared by the workbook and the specification document. Inputs come f                                        |
| <code>bind_cancel_field.py</code>  | Adds ONE field binding, <code>IsAdminChange</code> (string), to the cancellation endorsement object (id 300224988) for the Uponly TECH products only. It never touc                           |
| <code>build_api_docs.py</code>     | External API documentation (Markdown) + OpenAPI 3.0 spec for the Uponly insurance APIs. Internal details (tenant tables, configuration keys, code name                                        |
| <code>build_mapping_xlsx.py</code> | Workbook 'our APIs vs platform APIs': per our API the platform calls (SDK operation or REST) and the platform request. Built from the Groovy source, U                                        |
| <code>build_policy_model.py</code> | Builds <code>Afillar_PolicyModel_v0.4.xlsx</code> in the layout of the PA policy model workbook (v0.1), extended to all Uponly products and APIs.                                             |
| <code>build_product.py</code>      | Component-path fields are created as temporary ( <code>IsTemp=Y</code> , not stored). Make them persisted like Uponly's.                                                                      |
| <code>carrier_check.py</code>      | One flow per product (first variant, first plan): <code>CarrierCode</code> / <code>CarrierName</code> must be returned by every API. Writes <code>carrier/results.json</code> .               |
| <code>carrier_setup.py</code>      | Creates the insurer table <code>UP_Carrier</code> and the rows that expose <code>CarrierCode</code> / <code>CarrierName</code> (field rules, feed columns, master table list). Idempotent: id |
| <code>code_facts.py</code>         | Facts extracted from the real source files, tables and scripts, so that the developer guide cannot drift from the code.                                                                       |
| <code>dev12_sections.py</code>     | Developer guide v1.2: the content added after v1.1 (handover risks, workflow / roles / dashboard, notifications, portal, PA001 and iHub, runbook). Eac                                        |

| Script                   | Purpose (first docstring line)                                                                                                                         |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| dev_content.py           | Developer guide content (block list). Reference sections are generated from the real source, tables and scripts.                                       |
| dev_content12.py         | Developer guide v1.2 content (block list). Reference sections are generated from the real source, tables and scripts; sections 1.4, 1.5 and 15 to 19 a |
| fix_endo_fields.py       | Admin script: makes the platform's cancellation calculation pass for an Uponly product by binding the fields it reports as 'not bound'. usage: fix_end |
| make_plan_collections.py | One Postman collection per plan; one folder per API; real flow request/response + recorded failure examples.                                           |
| make_ref_postman.py      | Builds the TrialX collection in exactly the structure of the UPONLY_UAT reference collection.                                                          |
| make_rules.py            | Creates UP_FieldRule, UP_ApiConfig, UP_DocWording, UP_BenefitText in Uponly and loads their rows.                                                      |
| make_sohar_collection.py | Single 'Sohar Bank' collection: one example per product, token mapped, values parameterized.                                                           |
| make_tables.py           | Create the UP_ data tables in Uponly (definitions and rows copied from the tested TrialX tables).                                                      |
| make_templates2.py       | Bilingual (english, arabic) paragraphs from the long text rows of the application docx.                                                                |
| md_to_docx.py            | Minimal Markdown -> docx (headings, paragraphs, bullets, tables, bold, code spans). usage: md_to_docx.py in.md out.docx                                |
| model_data.py            | Collects the facts for the policy model workbook and the external API documentation from the build folder (product tables, field rules, code tables, s |
| os_gen.py                | usage: os_gen.py TEMPLATE_NAME html_file data.json out.pdf (uploads and generates)                                                                     |
| rating_build.py          | Create <CODE> PREM_CALC (copy of Uponly PA001_PREM_CALC) bound to a market product in the current tenant. usage: rating_build.py <PRODUCTCODE> [--conf |
| ref_flow.py              | Runs the PA001-style 11 step flow for every plan of every variant, using the request bodies of the UPONLY_UAT reference collection as the template, an |
| refresh_devdata.py       | Rebuilds devdata/assets.json (server metadata saved by the last push of every asset) and devdata/tables.json (v1.1 data plus the rows written by workf |
| render_docx.py           | DOCX version of the specification (same content blocks): real heading styles, TOC field, repeating table header rows, page numbers.                    |
| render_spec.py           | Renders the specification content to PDF (HTML + Chromium, TOC page numbers from a first pass) and DOCX (python-docx), in the layout of the InsureMO r |
| role_check.py            | Role / maker-checker / notification checks on the live MC tenant. One account holds all three roles in UP_UserRole, so roles are switched by (de)activ |
| run_tests.py             | Stored copy: large responses are cut to keep the result files small (the live copy keeps the full body for the follow-up steps).                       |
| security_probe.py        | Live security (VAPT style) probe of the deployed Uponly APIs in the test tenant. Writes security/results.json. Each check: id, category, title, what w |
| setcfg.py                | usage: setcfg.py Key Value [Key Value ...] - changes UP_ApiConfig values (record-save) and prints the old values.                                      |
| snapshot.py              | Read-only snapshot of PA001 related Uponly objects. usage: snapshot.py before after                                                                    |
| spec_content.py          | Content model of the API specification (list of blocks) in the structure of the InsureMO reference specification.                                      |
| split_util.py            | Splits UPDataUtil into focused functions and rewrites the callers. Usage: split_util.py <src-root> <out-root>                                          |
| stage_a_rows.py          | Stage A: new configuration rows (no defaults in code) + UP_FeedColumn.                                                                                 |
| stage_cd_rows.py         | Stages C + D: ChannelCode rules on every API, cancel-check / cancel-approve rules and config keys, ChannelCode feed column.                            |
| status_cancel_spike.py   | Spike: can an issued Uponly policy be set to Cancelled through the platform's own proposal update calls? Works on ONE throwaway Uponly policy only (th |
| uat_content.py           | UAT deployment plan content (block list) + data for the checklist workbook.                                                                            |
| uptab.py                 | Small helpers to create / extend Uponly data tables (same approach as make_rules.py).                                                                  |

| Script            | Purpose (first docstring line)                                                                                                                         |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| workflow_setup.py | Maker / Checker workflow set-up: creates UP_UserRole and UP_NotifyTemplate and the config keys / field rules the new APIs need. Idempotent for tables; |

## 12. Test Procedure for a Change

1. Before: `snapshot.py before` (PA001 guard) and copy `ic` to a backup folder.
2. Make the change; push with `imo icomposer push current` (the compile check must say `Compiled: yes`).
3. Run the affected cases (`run_tests.py all <filter>`) and, for response changes, `ref_flow.py` for one plan per product and `carrier_check.py`.
4. For anything touching input handling or errors run `security_probe.py`.
5. After: `snapshot.py after` and compare with `before` (PA001 and PA001\_TECH hashes identical).
6. Update the documents (specification, policy model, this guide) and the collection (`make_sohar_collection.py`), then run the collection in Newman.
7. Rollback: push the saved files back; for tables restore the saved row.

## 13. Environments and Deployment

The complete inventory, ordered steps, verification and rollback for deploying to UAT are in the separate document *Afillar API UAT Deployment Plan*. In short: 13 functions and 18 APIs (v1.2), 14 `UP_` tables with their rows (v1.2 added `UP_UserRole` and `UP_NotifyTemplate`), the product configuration on the platform, the output templates and the environment settings. The keys that differ per environment are `AllowedHosts`, `GatewayUrl` and `PrintUrl`; the ids of products, data tables and templates are tenant specific and must be re-resolved, not copied. **The UAT plan was written for v1.1: for v1.2 also deploy the portal page, the two new tables, the new keys and rules, set up users and roles, and the SNS accounts (section 19).**

## 14. Known Limitations and Backlog

| Item                                | Description                                                                                                                                                           | Suggested action                                                                |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Role / branch access                | Any authenticated user of a channel can call any API; the ChannelCode check isolates channels only.                                                                   | Access table per API and role (deferred by decision).                           |
| Security baselines in code          | UPSecurity constants with optional override keys.                                                                                                                     | Add the override keys / UP_FieldRule patterns and make a missing key UP-CONFIG. |
| Gateway findings                    | Wildcard CORS, no rate limiting, OPTIONS 500, attack-like query strings hold the connection.                                                                          | Platform team.                                                                  |
| ID number and TempData in responses | The ID number is returned in clear and a TempData block with encrypted copies is part of the platform body.                                                           | Decide on masking; strip TempData in UPShaper.                                  |
| Idempotency and audit               | No idempotency key; no audit record of who approved a cancellation beyond the platform endorsement.                                                                   | Idempotency key on create / approve; audit table.                               |
| Unit tests                          | Only black-box API tests exist.                                                                                                                                       | Data-driven tests for pure functions (refundCalc, entryErrors, scan).           |
| Per-plan collections                | 52 per-plan collections were not regenerated after the insurer fields were added (still valid; examples lack the two fields).                                         | Re-run make_plan_collections.py after ref_flow.py for all plans.                |
| PDF templates                       | The insurer name is not printed on the documents.                                                                                                                     | Add CarrierName to the print data and the templates.                            |
| v1.2 items                          | See the consolidated list in section 19.6 (notifications live, sign-in, portal URL, role checks on feed and master data, maker stamp, dashboard accuracy, documents). | Section 19.6.                                                                   |

### 14.1 Troubleshooting

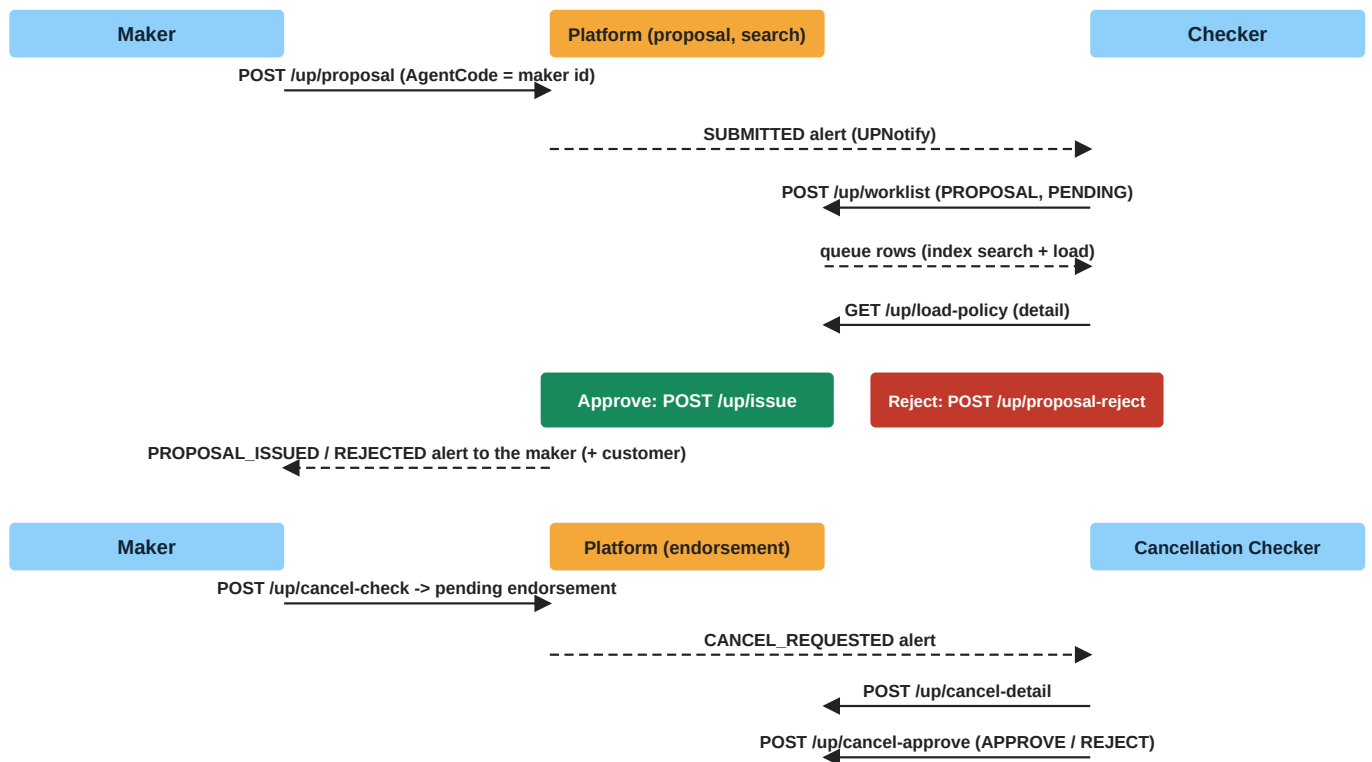
| Symptom                                     | Likely cause                                                                     | Check                                                                          |
|---------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| HTTP 500 UP-CONFIG                          | A configuration key, table row (carrier, channel, product) is missing            | Log of the trace id names the key / table                                      |
| Cancel approve returns 409 platform refused | Platform validation of the endorsement (for example OrgCode -1 on an old policy) | Text after the policy number; run probe_approve.py <PolicyNo> to see each step |
| Get Plans returns no plan                   | No active plan row for the variant and premium mode                              | UP_ProductPlanRelation                                                         |
| 400 UNKNOWN_VALUE on a code                 | Value missing in the code table                                                  | imo config codetable data --name X                                             |
| Push fails to compile                       | Callee not pushed yet or a Groovy static type error                              | Push order (section 3.4)                                                       |
| 401 from every call                         | Token expired                                                                    | Get a new token                                                                |
| Document not generated                      | Template missing or print host not in AllowedHosts                               | UP_DocTemplate, PrintUrl, AllowedHosts                                         |

### 14.2 First-Day Checklist for a New Developer

1. Install imo, log in to the MC tenant (imo auth login --profile portal:uponly).
2. Clone the source export, read sections 2 and 4 of this guide.
3. source env.sh, run carrier\_check.py (one flow per product) to see the APIs work.
4. Open UPCancelService.groovy and UPValidator.groovy next to sections 8 and 6.3.
5. Make a harmless change in a copy tenant or a branch: add a UP\_FieldRule row and see the 400.
6. Read the specification and the policy model workbook (API-Field sheets) for the contract.

# 15. Workflow, Roles and Dashboard

## 15.1 Overview



Rule: the person who created the proposal / request cannot approve or reject it (UPAccess.requireNotCreator, switch AllowSelfApproval).

Figure 3: Maker-checker flow for proposals and cancellations

Bank staff use the portal in three roles. The **Maker** creates quotations and proposals and raises cancellation requests. The **Proposal Checker** approves (issues) or rejects submitted proposals. The **Cancellation Checker** approves or rejects cancellation requests. No new workflow state is stored by our code: the platform already holds it. A proposal waiting for approval is a platform proposal with ProposalStatus 2. A cancellation request is the platform's pending cancellation endorsement (status 120). The queues are searches over that state, so there is nothing to keep in step and nothing to clean up.

## 15.2 Roles and Identity

The caller is taken from the platform token inside iComposer: `IComposerAppContext.getUserName()` and `IComposerAppContext.getIComposerCurrentUser()` (an `AppUser` with `userId`, `userName`, `realName`, `email`). The request body is never trusted for identity. The platform's own role list (`userRoleIds`) is empty for these users, so roles come from the table `UP_UserRole` (row per user and role; matched by user name, case-insensitive, or by platform user id). `UPAccess` offers:

| Method                                            | Behaviour                                                                                                                             |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <code>me()</code>                                 | UserName, UserId, DisplayName, Email, Branch, Roles[], RoleEnforcement, AllowSelfApproval. Roles is omitted from the JSON when empty. |
| <code>require(roles)</code>                       | Returns the caller; HTTP 403 UP-403 unless the caller holds one of the roles. Does nothing when RoleEnforcement is not Y.             |
| <code>requireAny()</code>                         | Any of the three roles (read access to policy data).                                                                                  |
| <code>requireNotCreator(caller, creatorId)</code> | Maker-checker rule: 403 when the caller created the item, unless AllowSelfApproval = Y or enforcement is off.                         |
| <code>requireCreator(caller, creatorId)</code>    | A maker may change only what they created (update proposal). An empty creator id (older proposals) is allowed.                        |
| <code>holdersOf(role) / contactOf(userId)</code>  | Recipients for alerts: active rows of a role, or the row of one user id.                                                              |

Role names are fixed strings used in four places and must stay in step: the config keys `RoleMaker`, `RoleProposalChecker`, `RoleCancellationChecker` (values `MAKER`, `PROPOSAL_CHECKER`, `CANCELLATION_CHECKER`), the `Role` column of `UP_UserRole`, the `Recipient` column of `UP_NotifyTemplate`, and the `ROLE` constants of the portal.

### Which API needs which role

| API                                                                                                            | Roles                                                                                    | Extra rule                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UPProposal (create)                                                                                            | Maker                                                                                    | Stamps the maker id on the proposal (15.3).                                                                                                                |
| UPUpdateProposal                                                                                               | Maker                                                                                    | Only the maker who created the proposal ( <code>requireCreator</code> ).                                                                                   |
| UPIssue                                                                                                        | Proposal Checker                                                                         | Not the creator of the proposal.                                                                                                                           |
| UPProposalReject                                                                                               | Proposal Checker                                                                         | Not the creator; a reason is required (field rule).                                                                                                        |
| UPCancelCheck                                                                                                  | Maker                                                                                    | Creates the pending request; alert to the Cancellation Checkers.                                                                                           |
| UPCancelDetail                                                                                                 | Maker or Cancellation Checker                                                            | Read only; <code>CanDecide</code> tells whether the caller may decide.                                                                                     |
| UPCancelApprove                                                                                                | Cancellation Checker                                                                     | Not the person who raised the request (checked inside <code>UPCancelService.decide</code> against the endorsement's <code>DataEntryUserId</code> ).        |
| UPWorklistApi                                                                                                  | PROPOSAL: Proposal Checker.<br>CANCELLATION: Cancellation Checker or Maker. MINE: Maker. | Type decides the role.                                                                                                                                     |
| UPDashboard, UPSHare, UPGetPlans, UPLoadPolicy, UPDocGeneration, UPApplicationDocGeneration, UPCommissionQuery | any of the three roles                                                                   | A caller who is only a Maker sees only their own figures in the dashboard.                                                                                 |
| UPMe                                                                                                           | none                                                                                     | Always answers; used by the portal to build the menu.                                                                                                      |
| UPListMasterTable, UPFeedFile                                                                                  | <b>no role check</b>                                                                     | Left open as before. Add <code>requireAny()</code> (or a checker role) to <code>UPFeedFile</code> if the feed must be restricted: it contains policy data. |

## 15.3 The Maker Stamp (AgentCode)

The platform search index cannot be filtered by the user who created a proposal. Of the indexed fields only `AgentCode` can carry the maker id. `UPProposal` therefore writes `AgentCode` = the caller's platform user id on every new proposal, and `UPUpdateProposal` keeps the stored value. `AgentCode` is in the server-owned field list of `UPSecurity` (`serverOwnedBaseline`), so a value sent by a caller is removed before the stamp is applied. The queue "My submissions" is a search with `AgentCode` = the caller's id; the maker-checker rule compares the same field with the checker's id. The id is mapped to a name through `UP_UserRole.UserId` (otherwise the platform's `DataEntryUserRealName` is shown).

*Consequences: proposals created before v1.2 have no maker id (they appear in the checker queue but never in "My submissions"); if the bank later needs real sales agent codes, another field and a new search condition are needed; the platform does not validate `AgentCode` against a code table (verified).*

## 15.4 Queues

### Proposal queue (type PROPOSAL and MINE)

- `UPWorklist.proposalList` runs `ProposalSdkClient...newQueryPolicyRequestBuilder` once per enabled product (the index supports only one exact product code per query), with `ProposalStatus` and the date window on `ProposalDate`, then merges the hits by date, takes the requested page and loads each proposal of the page (channel check, maker name, plan, premium).

- Status codes from `UP_ApiConfig`: `ProposalStatusPending` 2, `ProposalStatusIssued` 3, `ProposalStatusRejected` 4. Filter names `PENDING`, `ISSUED`, `REJECTED`, `ALL`.
- Limit: `PageSize` x `PageNo` <= `WorklistMaxRows` (1,000) and at most 100 hits per product per query. `WorklistPageSize` (20) is the default page size; the portal asks for 15.
- Fields returned per row: `ProposalNo`, `PolicyNo`, `ProductCode`, `ProductSubcode`, `ProductName`, `PlanId`, `PlanName`, `InsuredName`, `Premium`, `EffectiveDate`, `ProposalDate`, `Status`, `StatusCode`, `RejectReason` (`ProposalRejectDesc`), `Maker`, `MakerId`, `CarrierCode`, `CarrierName`.

### Cancellation queue (type CANCELLATION)

- `UPWorklist.endoSearch` calls the endorsement search `POST /platform/endo/v1/query` through `UPPlatformClient` (the same search service, endorsement index) with `EndoType` 3 (cancellation) and `EndoStatus`.
- `EndoStatusPending` 120 is a pending request, `EndoStatusIssued` 300 an approved cancellation, `EndoStatusWithdrawn` 600 a withdrawn one. A rejected request and a request replaced when an approval runs are both 600, so only `PENDING` and `ISSUED` can be listed.
- Rows add `RequestNo` (`EndoNo`), `RequestStatus`, `RequestedOn` and `CancellationDate`. Detail of one request is `UPCancelDetail`: it finds the pending endorsement with the signed policy id (as `pendingCancellation`) and returns reason (`OtherCancelReason`), requester (`DataEntryUserId` mapped to a name), the refund calculation and `CanDecide`.

## 15.5 Dashboard

`UPDashboard` has two sections so that the portal can run them in parallel. `SUMMARY` returns proposal counts (pending, issued, rejected, submitted, pending older than a day), cancellation counts and a per-product split: per product three policy searches (one per status), one for pending older than a day and two endorsement searches. `COMMISSION` takes one `ProductCode`, searches the issued policies by `EffectiveDate`, loads up to `DashboardMaxPolicies` of them and returns premium, commission, policies in force and cancelled (a policy whose status is not `PolicyStatusInforce` counts as cancelled), a by-plan table and the flags `IssuedInWindow`, `Counted`, `Truncated`. The portal calls `SUMMARY` once and `COMMISSION` for each of the 7 products at the same time (8 calls).

- Window: `FromDate` / `ToDate`, default the last 30 days, at most `DashboardMaxDays` (180) wide; otherwise 400.
- Scope: `ALL` or `MINE`. A caller who holds only the Maker role is always restricted to their own submissions (`AgentCode` condition).
- Cost: roughly 7 to 10 seconds per call in MC (about 40 platform calls for `SUMMARY`; one load per policy for `COMMISSION`).
- All figures include test data created in MC.

## 15.6 Maker-Checker Rule and the Test Switch

With `RoleEnforcement` = Y and `AllowSelfApproval` = N a user can never approve, reject or decide something they created. The live role check (`role_check.py`) proves this by switching `UP_UserRole` rows on and off for the single test account. With `AllowSelfApproval` = Y the rule is skipped, which is the state of MC so that one person can test the whole flow. The portal reads the switch from `/up/me` (`AllowSelfApproval`) and disables the approve and reject buttons only when the rule would refuse them.

## 15.7 New API Contracts (examples)

All requests are `POST` with a JSON body and the bearer token unless noted; `ChannelCode` is `S0HAR`. Errors use the standard body (section 9). The examples are shortened from real MC responses.

### GET /up/me

```
{ "UserName": "ravi.teja@insuremo.com", "UserId": "1188178494", "DisplayName": "Ravi Teja (test user - replace with bank staff)",
  "Email": "ravi.teja@insuremo.com", "Branch": "001",
  "Roles": ["MAKER", "PROPOSAL_CHECKER", "CANCELLATION_CHECKER"],
  "RoleEnforcement": "Y", "AllowSelfApproval": "Y" }
```

## POST /up/worklist

```
request : { "ChannelCode": "SOHAR", "Type": "PROPOSAL", "Status": "PENDING", "ProductCode": "HC001",
"FromDate": "2026-09-03", "ToDate": "2026-10-03", "PageNo": 1, "PageSize": 15 }
response: { "Type": "PROPOSAL", "Status": "PENDING", "PageNo": 1, "PageSize": 15, "Total": 117,
  "Records": [ { "ProposalNo": "PHC0010000000243", "PolicyNo": "", "ProductCode": "HC001", "ProductName":
"Hospital Cash Benefit",
  "PlanId": "1", "PlanName": "Plan 1", "InsuredName": "Workflow Test", "Premium": 70, "Maker": "Ravi Teja
(...)", "MakerId": "1188178494",
  "ProposalDate": "2026-10-03", "Status": "Bound", "StatusCode": "2", "RejectReason": "", "CarrierCode":
"AFIC", "CarrierName": "Afillar Insurance Company" } ] }
```

## POST /up/dashboard (SUMMARY and COMMISSION)

```
request : { "ChannelCode": "SOHAR", "Section": "SUMMARY", "FromDate": "2026-09-03", "ToDate": "2026-10-03",
"Scope": "ALL" }
response: { "Proposals": { "Pending": 567, "PendingOlderThanOneDay": 563, "Issued": 429, "Rejected": 332,
"Submitted": 1328 },
  "Cancellations": { "Pending": 110, "Issued": 199 }, "ByProduct": { "HC001": { "Pending": 117, "Issued": 74,
"Rejected": 51 } },
  "FromDate": "2026-09-03", "ToDate": "2026-10-03", "Scope": "ALL" }
request : { ..., "Section": "COMMISSION", "ProductCode": "HC001" }
response: { "ProductCode": "HC001", "IssuedInWindow": 69, "Counted": 47, "Truncated": true, "InForce": 16,
"Cancelled": 31,
  "Premium": 3080, "Commission": 2156, "CancelledPremium": 6510, "CancelledCommission": 4557,
  "ByPlan": [ { "PlanName": "Plan 1", "Policies": 8, "Premium": 560, "Commission": 392 } ] }
```

## POST /up/cancel-detail

```
request : { "ChannelCode": "SOHAR", "PolicyNo": "POUPPA00100000072" }
response: { "RequestNo": "POUPPA00100000072-001", "RequestStatus": "PENDING", "CancelReason": "spike",
"CancellationDate": "2026-10-04",
  "RequestedBy": "Ravi Teja (...)", "RequestedOn": "2026-10-03", "CanDecide": "Y", "CarrierCode": "AFIC",
"CarrierName": "Afillar Insurance Company",
  "PolicyInfo": { "PolicyNo": "...", "InsuredName": "...", "EffectiveDate": "...", "ExpiryDate": "..." },
"PolicyStatus": { "Code": "2", "Description": "Effective" },
  "PremiumDetails": { "Premium": 68, "FreeLookDays": 15, "DaysSinceCommencement": 0, "ProRataPremium": 68 },
  "RefundDetails": { "CancellationType": "FREE_LOOK", "RefundPercent": 100, "RefundPremium": 68 } }
```

## POST /up/share

```
request : { "ChannelCode": "SOHAR", "DocType": "POLICY", "Channel": "BOTH", "PolicyNo": "POHC00100000068",
"Mobile": "96891234567" }
  (DocType QUOTATION needs "Details": { InsuredName, ProductName, PlanName, Premium, EffectiveDate }
plus ProductCode / ProductSubcode and Email or Mobile)
response: { "DocType": "POLICY", "Channel": "BOTH", "Notifications": [ { "Event": "SHARE_POLICY", "Recipient":
"CUSTOMER", "Channel": "EMAIL", "To": "M***@gmail.com", "Status": "DRYRUN" },
  { "Event": "SHARE_POLICY", "Recipient": "CUSTOMER", "Channel": "SMS", "To": "****4567", "Status": "DRYRUN" } ]
}
```

Existing responses gained one additive field, `Notifications[]`, on `UPProposal (create)`, `UPIssue`, `UPProposalReject`, `UPCancelCheck` (only when a new request is created) and `UPCancelApprove`. A 403 body has `status 403` and `code UP-403`.

## 16. Notifications (Email and SMS)

### 16.1 Design

`UPNotify.fire(event, params, customer, makerUserId, onlyChannel)` reads every active row of `UP_NotifyTemplate` for the event, resolves the recipients and sends or records each message. It never throws: a problem becomes a result row, so a failed alert can not fail a proposal, an issue or a cancellation. The caller adds the result list to the response as `Notifications[]`. Recipients: a role (all active holders in `UP_UserRole`), `MAKER` (the creator of the item, looked up by platform user id) or `CUSTOMER` (the Email and Mobile of the customer, taken from the stored proposal, unmasked, or from the request of `UPShare`).

### 16.2 Events and Templates

| Event                                               | Recipient (email and SMS each) | Sent by                                                                                                                          |
|-----------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| SUBMITTED                                           | Proposal Checkers              | UPProposal after a successful create                                                                                             |
| PROPOSAL_ISSUED                                     | Maker, customer                | UPIssue                                                                                                                          |
| PROPOSAL_REJECTED                                   | Maker                          | UPProposalReject (placeholder {Reason})                                                                                          |
| CANCEL_REQUESTED                                    | Cancellation Checkers          | UPCancelCheck, only when a new request is created                                                                                |
| CANCEL_APPROVED                                     | Maker, customer                | UPCancelApprove (placeholder {Refund})                                                                                           |
| CANCEL_REJECTED                                     | Maker                          | UPCancelApprove with REJECT                                                                                                      |
| SHARE_QUOTATION,<br>SHARE_PROPOSAL,<br>SHARE_POLICY | Customer                       | UPShare (event name from the keys <code>ShareDocQuotation</code> , <code>ShareDocProposal</code> , <code>ShareDocPolicy</code> ) |

Placeholders (`{Name}`; a placeholder without a value becomes empty): `ProposalNo`, `PolicyNo`, `Insured`, `Product`, `Plan`, `Premium`, `EffectiveDate`, `Link` (`PortalUrl`), `Insurer`, `Checker`, `Maker` (both are the calling user's display name), `RequestNo`, `Reason`, `Refund`. `UPNotify.paramsOf(policy, extra)` builds them. A template row is identified by `TemplateKey = Event|Recipient|Channel` and has `Subject`, `Body`, `SmsTemplateCode` and `IsActive`.

### 16.3 Modes and Result Status

| Status          | Meaning                                                                                |
|-----------------|----------------------------------------------------------------------------------------|
| SENT            | The SNS call was made (NotifyMode LIVE).                                               |
| DRYRUN          | NotifyMode is not LIVE: nothing was sent. This is the state of MC.                     |
| NO_ADDRESS      | No email or mobile known for that recipient (for example the test user has no mobile). |
| NO_SMS_TEMPLATE | LIVE mode, SMS, and <code>SmsTemplateCode</code> is NONE.                              |
| FAILED          | The SNS call raised an exception (logged as <code>notification failed</code> ).        |

The `To` value in a result is masked (`r*@insuremo.com, *4567`).

### 16.4 SNS Integration Details

- Email: `SnsSdkClient.emailApi().newSendEmailRequestBuilder().requestBody(SendEmailRequest)`. Model fields (found by reflection in MC): `accountName`, `alias`, `attachments`, `bcc`, `cc`, `content`, `contentAlternative`, `contentType`, `customHeaders`, `replyTo`, `senderParams`, `subject`, `templateCode`, `templateParams`, `to`. The code fills `to`, `subject`, `content` and `accountName` (when `EmailAccountName` is not NONE).
- SMS: `smsApi().newSendSmsRequestBuilder().requestBody(SendSMSRequest)`. Model fields: `accountName`, `autoAddSign`, `senderParams`, `signName`, `templateCode`, `templateParams`, `to`, `useZhBracket`. **There is no free-text field**: the SMS is the SNS template named by `templateCode` filled with `templateParams`. The text in `UP_NotifyTemplate.Body` for SMS is therefore documentation and dry-run preview only; the real text lives in the SNS SMS template.
- The request model is built with `IComposerJsonUtils.mapToObject(map, Class)` from the Java field names. Whether the SNS service accepts exactly these names at runtime is unproven.

- MC findings: 0 email accounts and 0 SMS accounts. Email senders offered by the tenant: SENDGRID, SMTP, AWS PINPOINT, AWS SES, Microsoft Graph. SMS senders include AWS SNS, AWS End User Messaging, ALI, OneWay, EngageLab, worldSMS and several country specific ones.

## 16.5 Switching to Live Delivery

1. A tenant administrator creates an email account and an SMS account in the SNS service and, for SMS, one SMS template per message (the template parameters are the placeholder names above).
2. Put the account names in `EmailAccountName` / `SmsAccountName` (NONE means the default account) and each SMS template code in the `SmsTemplateCode` column of the SMS rows of `UP_NotifyTemplate` (record-save).
3. Check that every staff row in `UP_UserRole` has a real Email and Mobile.
4. Set `NotifyMode = LIVE` (`python3 setcfg.py NotifyMode LIVE`).
5. Exercise each event once (`role_check.py` produces all of them) and read the platform log for `notification failed` lines. Expect to adjust field names or template parameters on the first run.

## 17. The Portal (UIC Page)

### 17.1 What and Where

| Item     | Value                                                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Page     | Sohar Insurance Portal, path <code>sohar-insurance-portal</code> , page id 894239494239121408, tenant <code>uponly</code> , environment <code>portal</code>                                                                                                           |
| URL      | <code>https://portal.insuremo.com/uic-web/#/p/sohar-insurance-portal</code> (a one-time login-free link comes from <code>imo uic page exchange-code --path sohar-insurance-portal</code> )                                                                            |
| Source   | <code>uic_pages/uponly/portal/sohar-insurance-portal/index.html</code> (one self-contained HTML file with inline CSS and JavaScript; copy in <code>deliverables/Sohar_Insurance_Portal_index.html</code> ; also <code>metadata.json</code> , <code>prompt.md</code> ) |
| Template | standard UIC template (vanilla HTML and JavaScript, no build step, no external libraries; the charts are inline SVG)                                                                                                                                                  |
| Size     | about 750 lines; section 17.2 lists the functions                                                                                                                                                                                                                     |

| View (role)                                 | Screen                                                                                                    | APIs called                                                                                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dashboard (all)                             | Tiles, three charts, tables by product and plan; period and scope filter                                  | <code>/up/dashboard</code> (SUMMARY + 7 x COMMISSION)                                                                                                                     |
| New quotation (Maker)                       | Step 1 quotation (basic details, plan cards, share), step 2 proposal details, step 3 submitted            | <code>/up/get-plans</code> , <code>/up/share</code> , <code>/up/proposal</code> , <code>/up/doc_generation</code>                                                         |
| My submissions (Maker)                      | Queue MINE with filters; detail with reject reason and PDF / share                                        | <code>/up/worklist</code> (MINE), <code>/up/load-policy</code>                                                                                                            |
| Proposal approvals (Proposal Checker)       | Queue PROPOSAL; detail with approve / reject (reason in a dialog), PDF, share                             | <code>/up/worklist</code> , <code>/up/load-policy</code> , <code>/up/issue</code> , <code>/up/proposal-reject</code> , <code>/up/share</code>                             |
| Cancellations (Maker, Cancellation Checker) | Tab Requests (queue and detail with refund calculation and approve / reject), tab Raise a request (Maker) | <code>/up/worklist</code> (CANCELLATION), <code>/up/cancel-detail</code> , <code>/up/cancel-check</code> , <code>/up/cancel-approve</code> , <code>/up/load-policy</code> |
| Policies (all)                              | Search by policy or proposal number; certificate / application PDF, commission, share                     | <code>/up/load-policy</code> , <code>/up/doc_generation</code> , <code>/up/commission-query</code> , <code>/up/share</code>                                               |

### 17.2 Structure of the File

The script is organised in blocks, in this order: UIC helpers (token, tenant, URL), configuration (`CFG`, `ROLE`), utilities (`api()`, error rendering, modal, alerts), state and master data (`loadMaster`: `ProductMaster`, `ProductRule`, `PremiumMode`, code tables, `UP_Carrier` through `/up/list-master-table`), navigation by role (`NAV`, `buildNav`, `go`), shared renderers (`policySummary`, `getDoc`, `shareDialog`), the generic `Queue` class used by three views, proposal and cancellation detail, policies, the quotation wizard, the dashboard (inline SVG helpers `barSvg`, `hbarSvg`, `donutSvg`) and the wire-up (`bind`, `startApp`, sign-in / sign-out, `init`).

#### Function index of `index.html` (generated)

| Function                     | Parameters           | Line | Comment |
|------------------------------|----------------------|------|---------|
| <code>getHostname</code>     |                      | 252  |         |
| <code>getTenantCode</code>   |                      | 253  |         |
| <code>buildApiUrl</code>     | <code>apiPath</code> | 254  |         |
| <code>getAccessToken</code>  |                      | 256  |         |
| <code>portalToken</code>     |                      | 257  |         |
| <code>redirectToLogin</code> |                      | 258  |         |
| <code>apiUrl</code>          | <code>path</code>    | 267  |         |

| Function         | Parameters                       | Line | Comment                                                                                                        |
|------------------|----------------------------------|------|----------------------------------------------------------------------------------------------------------------|
| logCall          | method, path, body, status, resp | 279  |                                                                                                                |
| api              | method, path, body, quiet        | 285  | Calls an /up API. Returns {ok,status,data,blob}; never throws; 401 sends the user to login. quiet = no spinner |
| showAlert        | kind, html                       | 300  |                                                                                                                |
| errHtml          | res, fieldMap                    | 301  |                                                                                                                |
| showApiError     | res, fieldMap                    | 307  |                                                                                                                |
| download         | blob, name                       | 308  |                                                                                                                |
| openModal        | html                             | 309  |                                                                                                                |
| closeModal       |                                  | 310  |                                                                                                                |
| clearBad         | scope                            | 311  |                                                                                                                |
| fieldErr         | id, msg                          | 312  |                                                                                                                |
| notifyHtml       | list                             | 314  | Short text for the Email / SMS results returned by the APIs.                                                   |
| loadMaster       |                                  | 328  |                                                                                                                |
| products         |                                  | 334  |                                                                                                                |
| variantsOf       | code                             | 335  |                                                                                                                |
| buildNav         |                                  | 351  |                                                                                                                |
| go               | view                             | 360  |                                                                                                                |
| statusBadge      | r                                | 375  |                                                                                                                |
| coverRows        | plan                             | 376  |                                                                                                                |
| policySummary    | p                                | 378  | Read-only summary of a loaded proposal / policy.                                                               |
| getDoc           | type, no, subcode                | 387  |                                                                                                                |
| shareDialog      | ctx                              | 393  | Share dialog. ctx = {docType, proposalNo?, policyNo?, productCode?, subcode?, details?, email?, mobile?}       |
| Queue            |                                  | 416  |                                                                                                                |
| buildQueues      |                                  | 454  |                                                                                                                |
| openProposal     | row, mountId, canAct             | 461  | ---- proposal detail (Proposal Checker acts, Maker only views) ----                                            |
| openCancel       | row                              | 496  | ---- cancellation detail (Cancellation Checker acts) ----                                                      |
| cancelTab        | t                                | 516  |                                                                                                                |
| cancelLoadPolicy |                                  | 517  |                                                                                                                |
| cancelSubmit     |                                  | 524  |                                                                                                                |
| doSearch         |                                  | 538  |                                                                                                                |
| fillStatic       |                                  | 552  |                                                                                                                |
| onProduct        |                                  | 560  |                                                                                                                |
| onVariant        |                                  | 561  |                                                                                                                |
| termRequired     |                                  | 569  |                                                                                                                |
| onMode           |                                  | 570  |                                                                                                                |
| expiryOf         | effStr, term                     | 572  |                                                                                                                |
| ageAt            | dobStr, effStr                   | 573  |                                                                                                                |
| newStep          | n                                | 574  |                                                                                                                |
| validateQuote    |                                  | 578  |                                                                                                                |
| quoteBody        |                                  | 586  |                                                                                                                |
| doQuote          |                                  | 591  |                                                                                                                |

| Function         | Parameters | Line | Comment                      |
|------------------|------------|------|------------------------------|
| renderPlans      | plans      | 600  |                              |
| enterProposal    |            | 606  |                              |
| validateProposal |            | 607  |                              |
| proposalBody     |            | 614  |                              |
| doSubmitProposal |            | 625  |                              |
| resetWizard      |            | 636  |                              |
| barSvg           | rows, w    | 642  |                              |
| hbarSvg          | rows, w    | 647  |                              |
| donutSvg         | parts      | 652  |                              |
| dashWindow       |            | 660  |                              |
| loadDashboard    |            | 665  |                              |
| bind             |            | 699  |                              |
| startApp         |            | 707  |                              |
| loginMessage     | kind, text | 724  | ---- sign-in / sign-out ---- |
| endSession       | message    | 725  |                              |
| armIdle          |            | 727  |                              |
| doLogin          | e          | 728  |                              |
| init             |            | 742  |                              |

## 17.3 Sign-In and Session

- On load the page shows the sign-in screen. It offers a user name and password form and, when the browser already holds an InsureMO portal session (`sessionStorage.Authorization`), a second button "Continue as the user signed in to InsureMO".
- Sign-in: `POST {CFG.casUrl}/cas/ebao/v2/json/tickets` (same call as step 0 of the Postman collection) with headers `x-mo-user-source-id: platform`, `x-mo-tenant-id: {CFG.tenant}` and body `{username, password}`. The token is taken from the reply (`access_token`, `token`, `ticket`, `Token` or the same under `data`), as the collection's test script does. A failure shows one plain message ("Sign-in failed. Check your user name and password." or "not available") and no server text.
- Storage: the token is kept in `AUTH.token` and, for a reload, in `sessionStorage.sohar_token` (this tab only). The password is never stored or logged. Two modes: `login` (own sign-in) and `session` (the portal session token).
- Sign-out button and idle timeout (`CFG.idleMinutes = 30`) remove the stored token and reload the page. A 401 in `login` mode signs the user out with a message; in `session` mode it redirects to the InsureMO login as the UIC template does.
- After sign-in `/up/me` supplies the roles; no role gives an "Access denied" screen. The menu is only a convenience: every API enforces the role again.

*Not verified: a successful sign-in with a real password (no password was available), and whether the CAS ticket service always accepts a browser cross-origin call from the portal host (preflight returned `access-control-allow-origin: *`). Test with a real user first.*

## 17.4 API Access

`apiUrl(path) = buildApiUrl(CFG.routeBase) + path`, where `buildApiUrl` is the standard UIC helper: on a gateway host it uses the path as is, otherwise it prefixes `/api`; `{{tenant_code}}` is replaced (`platform` on the portal host). `CFG.routeBase` is `/{{tenant_code}}/api-orchestration-test/v1/flow`, the same route the collections use as `baseUrl`. All tests replaced the base with a stub through `sessionStorage.sohar_api_base`. That hook is still in the code (`API_OVERRIDE`): remove it before go-live, and confirm the real in-portal URL with a first manual call.

## 17.5 Configuration and Branding

- CFG (top of the script): `casUrl`, `tenant`, `idleMinutes`, `routeBase`, `channel` (SOHAR), `branch` (001), `userName` (SoharPortal, sent as `UserName` in quotations), `payLG` / `payLC` (payment info codes used by the reference flows), `riskCode` (R00001), `pageSize` (15).
- Products, variants, premium modes, age and term limits, free look and refund percent are read at start from the master data API (`UP_ProductMaster`, `UP_ProductRule`); nothing about a product is hard coded except `FEMALE_ONLY` = [FP001] (Female Protection is sold to women only).
- Marital status shows only the codes 1 and 2 of `InsMaritalStatus` because that code table also holds salutation values.
- Brand colours, font and chart colours are CSS variables in one `:root` block at the top (approximate Sohar International Bank public colours; the official guideline was not available). The logo is a text badge.

## 17.6 Deploy and Test the Portal

| Task                                            | Command / script                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deploy a change                                 | <code>imo uic page push --file uic_pages/uponly/portal/sohar-insurance-portal/index.html --memo "... " --profile portal:uponly</code> (pull first if someone else may have changed the page: <code>imo uic page pull --id 894239494239121408</code> )                                                                                                                                                                     |
| Get a login-free link                           | <code>imo uic page exchange-code --path sohar-insurance-portal --profile portal:uponly --json</code> (single use)                                                                                                                                                                                                                                                                                                         |
| Version history / rollback                      | <code>imo uic page history list --id 894239494239121408</code> , <code>imo uic page history rollback --page-id ... --version ...</code>                                                                                                                                                                                                                                                                                   |
| Browser tests (headless Chromium, stub gateway) | <code>flow2.js</code> (23 checks: dashboard, quote, share, submit, queue, approve, reject, cancellation, policy view), <code>login.js</code> (11 checks: sign-in screen), <code>run.js</code> (12-step flow per product); run with <code>node &lt;script&gt; &lt;outputDir&gt; &lt;ProductCode&gt;</code> ; screenshots in <code>deliverables/ui_test_workflow</code> , <code>ui_test_login</code> , <code>ui_test</code> |
| Role / notification API checks                  | <code>python3 role_check.py</code> (35 checks; toggles <code>UP_UserRole</code> rows and <code>AllowSelfApproval</code> , restores them at the end)                                                                                                                                                                                                                                                                       |

The browser tests need a valid `imo` token (they read it with `imo auth prepare`) and Playwright with Chromium. They talk to the real MC APIs and therefore create test proposals, policies and cancellations each time.

## 17.7 Portal Limitations

- A submitted proposal cannot be edited in the portal (the API supports it for the creating maker); a rejected proposal is resubmitted as a new quotation.
- The proposal detail shows what the platform returns: the mobile is masked by the platform; the ID number is masked by the page (last four characters).
- Queues show up to 15 rows per page; all totals include test data.
- The dashboard needs 7 to 10 seconds and is capped as described in 15.5.
- No audit trail beyond the platform's own records; no export.

## 18. PA001, iHub and the Other Assets in the Tenant

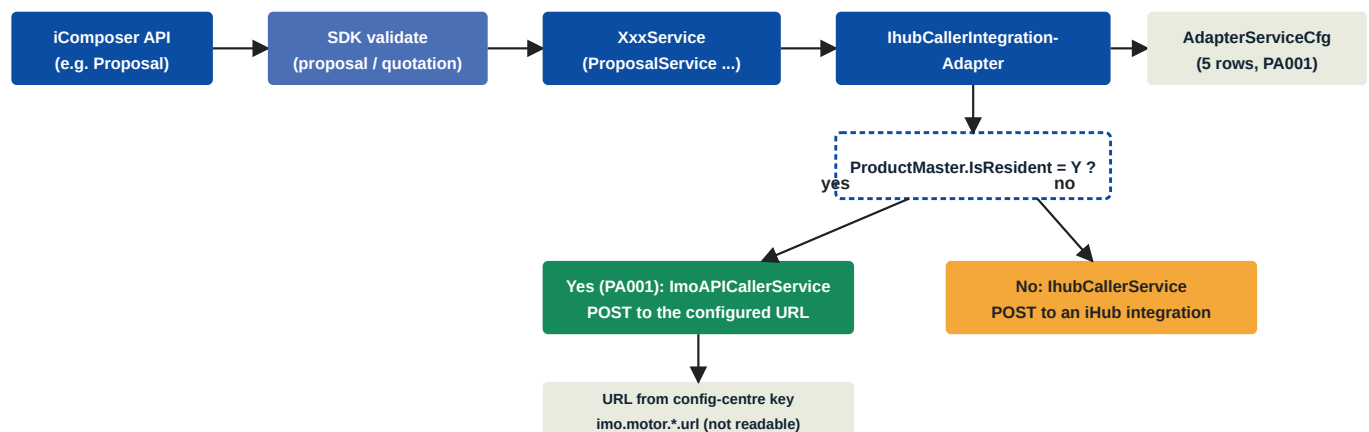
This section records what was found about the existing PA001 solution so that nobody assumes the new products work the same way. It is based on a read-only copy of the tenant's iComposer code and tables taken earlier in this project. A second, live check was not possible when this guide was written because the `imo` login had expired, so verify anything critical in the tenant.

### 18.1 What Exists

| Group (iComposer)                  | Content                                                                                                                                                                                                                                                                                                                                                                                              | Notes                                                                                                       |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Policy_API (group 572066178)       | 13 APIs: GetPlans, Quote, Proposal, UpdateProposal, IssuePolicyAPI, ProposalReject, LoadPolicyData, DocGeneration, ApplicationDocGeneration, FeedFile, QuickProposal, CreateUpdateProposal, CreateProposalReject. 8 functions: ProposalService, UpdateProposalService, IssueService, ProposalRejectService, QuoteService, ImoAPICallerService, APICallerUtils, CreateUpdateProposalBusinessFunction. | The PA001 flow of the reference collection (routes <code>/get-plans</code> , <code>/proposal</code> , ...). |
| POLICY_API (group 764483681)       | 17 APIs (quote, policy, claims, document and master-table APIs) and 33 functions, including the adapter functions <code>IhubCallerIntegrationAdapter</code> , <code>IhubTransformerCallerIntegrationAdapter</code> , <code>IhubCallerService</code> , <code>RestClientService</code> , <code>DataTableUtil</code> , <code>ExceptionHandler</code> , <code>ErrorCodeEnum</code> .                     | Shared building blocks of the PA001 design.                                                                 |
| UP_PRODUCTS_API (group 1188491119) | Everything described in this guide.                                                                                                                                                                                                                                                                                                                                                                  | Created for the new products.                                                                               |
| Platform AI group (706710450)      | About 45 <code>AI_*</code> APIs supplied by the platform.                                                                                                                                                                                                                                                                                                                                            | Not ours; do not change.                                                                                    |

### 18.2 How the PA001 APIs Work

PA001 APIs (group Policy\_API, routes `/get-plans`, `/proposal` ...)



New products (group UP\_PRODUCTS\_API, routes `/up/...`)



Figure 5: PA001 call pattern (top) compared with the new products (bottom)

1. The API validates the request with the platform SDK (`proposalApi().newValidateRequestBuilder()` or the quotation equivalent).
2. A service function (for example `ProposalService.getCarrierProposal()`) takes the provider code (`CarrierCode`, default AFIC), the country (OM) and the transaction type code (GETPLANS, PROPOSAL, ISSUEPOLICY, UPDATEPROPOSAL, PROPOSALREJECT) and looks the numeric transaction type up in the iTable `AdapterServiceCfg` (5 rows for PA001: 10001 to 10005).
3. `IhubCallerIntegrationAdapter.callIntegrationAdapter` builds an `IComposerIntegrationAdapterRequest` and calls the platform's integration adapter service. The adapter row names an implementation

(`iComposer.IhubCallerIntegrationAdapter`) and a URL setting (`imo.motor.quote.url`, `imo.motor.proposal.url`, `imo.motor.issuepolicy.url`, `imo.motor.updateproposal.url`, `imo.motor.proposalreject.url`) that is stored in the platform's config centre.

4. `sendRequest` then branches on `ProductMaster.IsResident` of the product: **Y** posts to the URL through `ImoAPICallerService` (an InsureMO API, no iHub); otherwise `IhubCallerService.executeIntegration` posts to an iHub integration and turns an iHub error into `IHUB_SERVICE_CALL_ERROR` / `IHUB_SERVICE_INTERNAL_ERROR`. For non-resident products the answer is also run through `CreateUpdateQuoteBusinessFunction`.
5. Both PA001 variants have `IsResident` = **Y** in the product table, so PA001 should take the direct branch.

## 18.3 Comparison

| Topic               | PA001 APIs                                                                                                                | New /up APIs                                                                                                                                              |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| iComposer group     | <code>Policy_API</code> and <code>POLICY_API</code>                                                                       | <code>UP_PRODUCTS_API</code>                                                                                                                              |
| Routes              | <code>/get-plans</code> , <code>/proposal</code> , <code>/issue</code> ...                                                | <code>/up/get-plans</code> , <code>/up/proposal</code> , <code>/up/issue</code> ...                                                                       |
| Product data tables | <code>ProductMaster</code> , <code>ProductPlanRelation</code> , <code>DocTemplate</code> , <code>AdapterServiceCfg</code> | <code>UP_ProductMaster</code> , <code>UP_ProductPlanRelation</code> , <code>UP_DocTemplate</code> , ... ( <code>UPDataUtil</code> maps the logical names) |
| Validation          | Platform SDK validate                                                                                                     | <code>UPValidator</code> with <code>UP_FieldRule</code> rows (plus the platform's own checks on save and issue)                                           |
| Backend call        | Integration adapter, URL from the config centre, direct IMO call for resident products, iHub for others                   | Platform SDK and endorsement REST called directly with the caller's token                                                                                 |
| iHub                | Available through the adapter (not used for resident products)                                                            | Not used                                                                                                                                                  |
| Settings            | Config centre keys and <code>AdapterServiceCfg</code>                                                                     | <code>UP_ApiConfig</code> (a missing key is <code>UP-CONFIG</code> )                                                                                      |
| Errors              | <code>ExceptionHandler</code> , <code>ErrorCodeEnum</code>                                                                | <code>UPErrorHandler</code> (400 / 403 / 404 / 409 / 500 contract)                                                                                        |

## 18.4 What Is Not Known

- Where the config-centre URLs `imo.motor.*.url` point (they are not readable with the tools used), and therefore what the PA001 "IMO API" branch calls.
- Whether any iHub integration, validator, transformer or Rego rule exists for PA001 or any other product in the tenant (the `imo ihub integration search` command was not run successfully).
- Whether the code copy used for this analysis still matches the tenant (taken earlier in the project).
- How PA001 behaves in UAT; the UAT plan lists its dependencies.

To close these points: log in again (`imo auth login --profile portal:upononly`), run `imo ihub integration search --scope own --profile portal:upononly --json`, reload the `Policy_API` and `POLICY_API` groups into a scratch workspace (`imo icomposer init --group-id <id>`) and read the config centre values with a platform administrator. Do not change PA001: run `snapshot.py` before and after around any work in the same tenant.

## 18.5 If iHub Is Wanted Later

A bank or carrier integration for the new products would be a new function beside `UPPlatformClient` that posts the request to an iHub integration URL held in `UP_ApiConfig` (with the allowed host list applied as `requireAllowedHost` already does), a new key per transaction type, and a switch per product in `UP_ProductRule`. The PA001 adapter (`IhubCallerIntegrationAdapter` / `AdapterServiceCfg`) can be studied but should not be shared: it is wired to PA001's rows and to `ProductMaster`.

# 19. Operations Runbook and Handover Checklist

## 19.1 Switches and How to Change Them

| Key (UP_ApiConfig)                           | MC now                                                         | Meaning                                                | Go-live value             |
|----------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|---------------------------|
| RoleEnforcement                              | Y                                                              | Role checks active. N = every API open as before v1.2. | Y                         |
| AllowSelfApproval                            | Y                                                              | Creator may approve own item (single test account).    | N                         |
| NotifyMode                                   | DRYRUN                                                         | LIVE sends through SNS, anything else records only.    | LIVE after accounts exist |
| EmailAccountName / SmsAccountName            | NONE                                                           | SNS account names (NONE = default account).            | per tenant                |
| PortalUrl                                    | https://portal.insuremo.com/uic-web/#/p/sohar-insurance-portal | Link placed in messages.                               | per environment           |
| DashboardMaxDays, DashboardMaxPolicies       | 180, 60                                                        | Window and per-product policy cap.                     | tune                      |
| WorklistPageSize, WorklistMaxRows            | 20, 1000                                                       | Default page size; index limit.                        | keep                      |
| EndoSearchPath, SearchPath, EndoSearchModule | set                                                            | Platform search routes used by the queues.             | check per tenant          |

Change a key: `python3 setcfg.py Key Value [Key Value ...]` (it reads `UP_ApiConfig` with `record-list`, saves the row with `record-save` and prints the old value). Changes apply from the next request; no deployment is needed.

## 19.2 Users and Roles

1. A platform administrator creates the person's InsureMO user in the tenant (this solution cannot create users or passwords).
2. Add one row per role to `UP_UserRole` (`record-batch-save`; `UserRoleKey = user|ROLE` must be unique, `Id` unique): `UserName` (as the platform reports it, usually the email), `UserId` (platform user id, needed for alerts to the maker), `Role`, `DisplayName`, `Branch`, `Email`, `Mobile`, `IsActive` Y.
3. To remove access set `IsActive` to N (`record-save`) rather than deleting the row.
4. Check with the user's own login: the portal header shows name and roles; `GET /up/me` returns the same.

`workflow_setup.py` shows the row format. Re-running the setup scripts is not idempotent for rows: the platform refuses a duplicate `Id`.

## 19.3 Troubleshooting the v1.2 Features

| Symptom                               | Likely cause                                                                                      | Check                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 403 UP-403 for everything             | User not in <code>UP_UserRole</code> or the row is inactive                                       | <code>GET /up/me</code> ; the table rows                                      |
| 403 on approve / reject               | Maker-checker rule (creator = caller) with <code>AllowSelfApproval = N</code>                     | <code>cancel-detail</code> shows <code>CanDecide</code> ; use another checker |
| Queue is empty but proposals exist    | Date window, product filter, or search index delay of a few seconds after a change                | Retry; check <code>ProposalStatus</code> of the proposal                      |
| "My submissions" misses old proposals | Created before v1.2 (no <code>AgentCode</code> )                                                  | Expected                                                                      |
| Dashboard slow or truncated           | Cap <code>DashboardMaxPolicies</code> ; many policies                                             | Narrow the window; raise the cap carefully                                    |
| Notifications show DRYRUN             | <code>NotifyMode</code> is not LIVE                                                               | Section 16.5                                                                  |
| <code>NO_ADDRESS</code> for the maker | The maker has no row ( <code>UserId</code> ) or no <code>Email</code> in <code>UP_UserRole</code> | Add the user row                                                              |
| Portal: "Access denied"               | User has no role                                                                                  | Add roles (19.2)                                                              |

| Symptom                                     | Likely cause                                                             | Check                                                                |
|---------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------|
| Portal: first call returns 404              | Route prefix of the real portal frame                                    | Section 17.4; browser network tab                                    |
| Sign-in always fails                        | Wrong credentials, user in another tenant, or reply field name different | Browser network tab for the ticket call; compare with Postman step 0 |
| Push fails with token is invalid or expired | imo login expired (24 hours)                                             | imo auth login --profile portal:uponly                               |

## 19.4 Go-Live Checklist for the v1.2 Additions

1. Replace the test rows of `UP_UserRole` with bank staff; set `AllowSelfApproval = N`; run `role_check.py` (it flips the switch temporarily: run it in a test tenant, or accept the brief change).
2. Create SNS email and SMS accounts and templates; fill the account keys and `SmsTemplateCode`; set `NotifyMode = LIVE`; send one test of every event.
3. Delete the probe API `UPSpokeWho`; remove the `sohar_api_base` hook (`API_OVERRIDE`) from the portal; confirm the in-portal route with a manual call.
4. Verify a real password sign-in (field name of the CAS reply).
5. Decide about role checks on `UPFeedFile` and `UPListMasterTable`; decide about masking the ID number and the platform `TempData` block.
6. Update the specification, UAT plan and checklist, policy model, OpenAPI and Postman collection for the new APIs, tables and keys.
7. Re-run the full suites: `run_tests.py`, `carrier_check.py`, `role_check.py`, the three browser scripts, `security_probe.py`, `snapshot.py` before / after.
8. Decide what to do with the test data in the tenant (it cannot be deleted through the APIs).

## 19.5 What a New Developer Needs

| Need             | Detail                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Access           | An InsureMO user in the MC tenant with iComposer, iTables, product configuration and UIC rights; the imo CLI logged in ( <code>imo auth login --profile portal:uponly</code> ). No passwords or tokens are stored in the repository or in this guide.                                                                                                                                                                         |
| Repository       | Branch <code>claude/wizardly-newton-ivfnle</code> of <code>br8innovates/Tablet</code> . Portal: <code>uic_pages/</code> . Delivered copies and tests: <code>deliverables/</code> ( <code>workflow_src/ic_src</code> holds the iComposer source of v1.2 and the set-up and test scripts; <code>ui_test*</code> hold the screenshots). The full build folder (document generators, test suites, samples) is in the package zip. |
| Order of reading | Section 1.4, then 2, 15, 17, 18; then 4 (code reference) next to the source; then 6 (configuration) and 19.                                                                                                                                                                                                                                                                                                                   |
| First actions    | <code>GET /up/me</code> ; run <code>role_check.py</code> (several minutes); open the portal and walk the flow in all three views; read <code>UPAccess</code> , <code>UPWorklist</code> , <code>UPNotify</code> (about 450 lines together).                                                                                                                                                                                    |

## 19.6 Backlog (Consolidated)

| # | Item                                                                      | Suggested action                                                              |
|---|---------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1 | Notifications never exercised LIVE                                        | Section 16.5                                                                  |
| 2 | Real password sign-in untested; portal URL in the real frame untested     | Manual verification, then remove <code>sohar_api_base</code>                  |
| 3 | No role check on <code>UPFeedFile</code> / <code>UPListMasterTable</code> | Add <code>requireAny()</code> or a checker role                               |
| 4 | Maker id in <code>AgentCode</code>                                        | Dedicated field or index field when the platform allows                       |
| 5 | Dashboard accuracy and speed (cap 60, 7 to 10 s)                          | Pre-aggregation (batch) or a reporting table if figures must be exact         |
| 6 | Rejected cancellation requests not listable                               | Record the decision outside the platform or read endorsement history          |
| 7 | No audit table (who approved what, when)                                  | Audit table or the platform audit module                                      |
| 8 | ID number and <code>TempData</code> in responses                          | Mask / strip in <code>UPShaper</code>                                         |
| 9 | Edit of a submitted proposal in the portal                                | Add an edit view (fields are masked in the load response: ask for them again) |

| #  | Item                                                                                                                            | Suggested action                                                                                         |
|----|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 10 | Documents lag behind the code                                                                                                   | Update specification, plan, model, OpenAPI, collection                                                   |
| 11 | PA001 / iHub open questions (18.4)                                                                                              | Live check after login                                                                                   |
| 12 | Unit tests for pure functions                                                                                                   | Data-driven tests                                                                                        |
| 13 | Wrong ID type and marital status codes are accepted by Get Plans (neither the platform validation nor UP_FieldRule checks them) | Add UP_FieldRule rows with CodeTable IdType / InsMaritalStatus (note the odd values in InsMaritalStatus) |

## Appendix A. File Inventory

| File (under ic/)                                                                                | Lines | Status   |
|-------------------------------------------------------------------------------------------------|-------|----------|
| src/dev/Tenant/UP_PRODUCTS_API/api/UPApplicationDocGeneration/UPApplicationDocGeneration.groovy | 25    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPCancelApprove/UPCancelApprove.groovy                       | 25    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPCancelCheck/UPCancelCheck.groovy                           | 26    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPCancelDetail/UPCancelDetail.groovy                         | 28    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPCancelPolicy/UPCancelPolicy.groovy                         | 5     | inactive |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPCommissionQuery/UPCommissionQuery.groovy                   | 137   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPDashboard/UPDashboard.groovy                               | 50    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPDocGeneration/UPDocGeneration.groovy                       | 24    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPFeedFile/UPFeedFile.groovy                                 | 178   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPGetPlans/UPGetPlans.groovy                                 | 114   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPIssue/UPIssue.groovy                                       | 58    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPListMasterTable/UPListMasterTable.groovy                   | 37    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPLoadPolicy/UPLoadPolicy.groovy                             | 37    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPMe/UPMe.groovy                                             | 13    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPProposal/UPProposal.groovy                                 | 44    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPProposalReject/UPProposalReject.groovy                     | 50    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPShare/UPShare.groovy                                       | 72    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPSpikeWho/UPSpikeWho.groovy                                 | 20    | inactive |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPUpdateProposal/UPUpdateProposal.groovy                     | 62    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/api/UPWorklistApi/UPWorklistApi.groovy                           | 35    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPAccess/UPAccess.groovy                                | 124   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPCancelService/UPCancelService.groovy                  | 364   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPDataUtil/UPDataUtil.groovy                            | 235   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPDocBuilder/UPDocBuilder.groovy                        | 195   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPDocService/UPDocService.groovy                        | 105   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPErrorHandler/UPErrorHandler.groovy                    | 112   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPNotify/UPNotify.groovy                                | 166   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPPlatformClient/UPPlatformClient.groovy                | 49    | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPProductData/UPProductData.groovy                      | 213   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPSecurity/UPSecurity.groovy                            | 116   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPShaper/UPShaper.groovy                                | 126   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPValidator/UPValidator.groovy                          | 334   | active   |
| src/dev/Tenant/UP_PRODUCTS_API/function/UPWorklist/UPWorklist.groovy                            | 408   | active   |

## Appendix B. Table Definitions

### UP\_ApiConfig (id 1188575596, 113 rows)

| Column | Data type | Primary key |
|--------|-----------|-------------|
| Id     | String    | Y           |
| Api    | String    |             |
| Scope  | String    |             |
| Key    | String    |             |
| Value  | String    |             |

### UP\_FieldRule (id 1188575333, 167 rows)

| Column     | Data type      | Primary key |
|------------|----------------|-------------|
| Id         | String         | Y           |
| Api        | String         |             |
| Path       | String         |             |
| Label      | String         |             |
| Required   | String         |             |
| DataType   | String         |             |
| MinLength  | Integer (long) |             |
| MaxLength  | Integer (long) |             |
| Pattern    | String         |             |
| MinValue   | Decimal        |             |
| MaxValue   | Decimal        |             |
| CodeTable  | String         |             |
| CodeColumn | String         |             |
| EnumValues | String         |             |
| AppliesTo  | String         |             |
| WhenPath   | String         |             |
| WhenValue  | String         |             |

### UP\_ProductMaster (id 1188475672, 11 rows)

| Column             | Data type | Primary key |
|--------------------|-----------|-------------|
| Id                 | String    | Y           |
| Description        | String    |             |
| LOB                | String    |             |
| IMOProductCode     | String    |             |
| IsRenewable        | String    |             |
| IsActive           | String    |             |
| IsFlatRate         | String    |             |
| Premium            | Decimal   |             |
| SumInsured         | Decimal   |             |
| HasPlans           | String    |             |
| CarrierCode        | String    |             |
| CarrierProductCode | String    |             |
| CarrierProductName | String    |             |

| Column                   | Data type      | Primary key |
|--------------------------|----------------|-------------|
| IsResident               | String         |             |
| BenefitRelationTableName | String         |             |
| LobId                    | Integer (long) |             |
| CarrierName              | String         |             |

#### UP\_ProductPlanRelation (id 1188475676, 57 rows)

| Column             | Data type | Primary key |
|--------------------|-----------|-------------|
| Id                 | String    | Y           |
| IMOProductCode     | String    |             |
| IMOPlanCode        | String    |             |
| IMOPlanName        | String    |             |
| CarrierCode        | String    |             |
| CarrierProductCode | String    |             |
| CarrierProductName | String    |             |
| CarrierPlanCode    | String    |             |
| CarrierPlanName    | String    |             |
| IsActive           | String    |             |
| Premium            | Decimal   |             |
| SumInsured         | Decimal   |             |
| Commission         | Decimal   |             |
| PremiumModeCode    | String    |             |

#### UP\_PlanBenefitRelation (id 1188474890, 148 rows)

| Column              | Data type      | Primary key |
|---------------------|----------------|-------------|
| Id                  | String         | Y           |
| IMOProductCode      | String         |             |
| IMOPlanCode         | String         |             |
| CarrierCode         | String         |             |
| CarrierProductCode  | String         |             |
| CarrierPlanCode     | String         |             |
| BenefitCode         | String         |             |
| BenefitDesc         | String         |             |
| BenefitLevel        | String         |             |
| ParentBenefitCode   | String         |             |
| BenefitLimit        | String         |             |
| BenefitLimitDesc    | String         |             |
| IsMainBenefit       | String         |             |
| IsAdditionalBenefit | String         |             |
| Notes               | String         |             |
| IsApplicable        | String         |             |
| LimitBasis          | String         |             |
| MaxDays             | Integer (long) |             |

**UP\_ProductRule (id 1188475684, 11 rows)**

| Column               | Data type      | Primary key |
|----------------------|----------------|-------------|
| Id                   | String         | Y           |
| ProductCode          | String         |             |
| ProductSubcode       | String         |             |
| PremiumModes         | String         |             |
| MinTerm              | Integer (long) |             |
| MaxTerm              | Integer (long) |             |
| MinAge               | Integer (long) |             |
| MaxAge               | Integer (long) |             |
| MaxEntryAge          | Integer (long) |             |
| MaxExitAge           | Integer (long) |             |
| MandatoryDocuments   | String         |             |
| SinglePremiumRule    | String         |             |
| FreeLookDays         | Integer (long) |             |
| ProRataRefundPercent | Decimal        |             |

**UP\_DocTemplate (id 1188475689, 22 rows)**

| Column         | Data type | Primary key |
|----------------|-----------|-------------|
| Id             | String    | Y           |
| DocumentCode   | String    |             |
| TemplateName   | String    |             |
| FileName       | String    |             |
| ProductCode    | String    |             |
| ProductSubcode | String    |             |
| DocType        | String    |             |

**UP\_DocWording (id 1188556760, 27 rows)**

| Column   | Data type | Primary key |
|----------|-----------|-------------|
| Id       | String    | Y           |
| ModeCode | String    |             |
| Key      | String    |             |
| Value    | String    |             |

**UP\_BenefitText (id 1188575620, 21 rows)**

| Column      | Data type | Primary key |
|-------------|-----------|-------------|
| Id          | String    | Y           |
| ProductCode | String    |             |
| BenefitCode | String    |             |
| NameEn      | String    |             |
| NameAr      | String    |             |

**UP\_FeedColumn (id 1188591810, 29 rows)**

| Column   | Data type      | Primary key |
|----------|----------------|-------------|
| Id       | String         | Y           |
| Sequence | Integer (long) |             |

| Column  | Data type | Primary key |
|---------|-----------|-------------|
| Heading | String    |             |
| Source  | String    |             |

#### UP\_Channel (id 1188591925, 1 rows)

| Column      | Data type | Primary key |
|-------------|-----------|-------------|
| Id          | String    | Y           |
| ChannelCode | String    |             |
| Description | String    |             |
| IsActive    | String    |             |

#### UP\_Carrier (id 1188663015, 1 rows)

| Column      | Data type | Primary key |
|-------------|-----------|-------------|
| CarrierCode | String    | Y           |
| CarrierName | String    |             |
| IsActive    | String    |             |

#### UP\_UserRole (id 1188710773, 3 rows)

| Column      | Data type | Primary key |
|-------------|-----------|-------------|
| UserRoleKey | String    | Y           |
| UserName    | String    |             |
| UserId      | String    |             |
| Role        | String    |             |
| DisplayName | String    |             |
| Branch      | String    |             |
| Email       | String    |             |
| Mobile      | String    |             |
| IsActive    | String    |             |

#### UP\_NotifyTemplate (id 1188710781, 22 rows)

| Column          | Data type | Primary key |
|-----------------|-----------|-------------|
| TemplateKey     | String    | Y           |
| Event           | String    |             |
| Recipient       | String    |             |
| Channel         | String    |             |
| Subject         | String    |             |
| Body            | String    |             |
| SmsTemplateCode | String    |             |
| IsActive        | String    |             |